

**TRACE: TEXTUAL AND RELATIONAL ANALYSIS OF COMMUNITY
EXTREMISM IN BLM TWITTER USING BERT AND
GRAPH NEURAL NETWORK**

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ACRONYMS AND ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
BLM	Black Lives Matter
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
NLP	Natural Language Processing
PR-AUC	Precision-Recall Area Under the Curve

ABSTRACT

The rapid spread of extremist content on social media platforms such as X/Twitter poses significant societal risks by amplifying hate speech and divisive narratives within movements like Black Lives Matter (BLM). This study investigates the effectiveness of Graph Neural Networks (GNNs) in predicting engagement patterns between users and extremist tweets in BLM-related discussions. A dual-method approach was employed, combining the pre-trained Bidirectional Encoder Representations from Transformers (BERT) HateXplain model for semantic classification of extremist, offensive, and neutral tweets with GNN architectures—specifically Graph Convolutional Network (GCN) and GraphSAGE—for link prediction. Fine-tuning BERT surpassed the baseline performance of 68.6%, achieving an F1 score of 78.7%, indicating improved classification of extremist content. Ablation studies showed that GraphSAGE outperformed GCN, achieving an accuracy of 88.7%, F1 of 89.8%, and PR-AUC of 92.5%. Integrating the fine-tuned BERT dataset with GraphSAGE for link prediction yielded strong performance: accuracy 0.917 ± 0.003 , precision 0.862 ± 0.004 , recall 0.994 ± 0.002 , F1-score 0.923 ± 0.002 , and PR-AUC 0.905 ± 0.004 .

Future research should focus on improving detection for minority classes, leveraging semi-supervised learning, and developing uncertainty-aware link prediction. Incorporating temporal modeling and multimodal data could further enhance detection of evolving extremist activity. Overall, the study demonstrates the effectiveness of combining semantic and relational modeling for identifying extremist engagement, providing a foundation for robust and responsible monitoring of online radicalization.

Keywords: graph neural networks, bert, hatexplain, link prediction, extremism

Chapter 1 INTRODUCTION

1.1 Background of the Study

1.1.1 Social Media and Political Activism

Social media has transformed political discourse, evolving from networking platforms into powerful vehicles for activism (Tufekci, 2017). These platforms democratize participation by allowing users to create and share political content regardless of geography or affiliation. The tone and nature of discussions—whether supportive, critical, or hostile—shape how narratives emerge and circulate (Hawkins, 2023). This dynamic was clearly visible in the Black Lives Matter (BLM) movement, where X/Twitter amplified voices against police brutality, particularly following the murder of George Floyd. Since its inception in 2013, the hashtag #BlackLivesMatter has served as a crucial tool for organizing, mobilizing, and raising awareness.

BLM was selected as the focus of this study because it stands as one of the most significant social justice movements of the 21st century, generating extensive online discourse, especially on X/Twitter. Its digital presence has been marked by unprecedented scale and engagement, with the hashtag used over 44 million times in the period following George Floyd's death in 2020 (Atske & Atske, 2023). As a decade-long movement that has fundamentally shaped online activism and political dialogue, BLM offers a compelling case for investigating how extremist content spreads within politically charged discussions.

Research further shows that BLM's prominence on social media has made it a frequent target for counter-narratives and extremist messaging from opposing groups (Freelon et al., 2016; Gallagher et al., 2021). Because BLM has remained highly visible online for many years and across major events, it offers valuable long-term data for studying how online extremist content first appears and then spreads (Bonilla & Rosa, 2015).

1.1.2 The Spread of Extremist Content Online

Research shows social media design can amplify extremist content online. On X/Twitter, recommendation algorithms tend to connect new users with celebrities, influencers, and political figures aligned with their interests, forming weak ties that may increase exposure to radical ideas and contribute to echo chambers (Wolfowicz et al., 2021; Torregrosa et al., 2020). These systems exploit psychological factors, including confirmation bias, social validation, and emotional arousal (responding to content that triggers strong emotional responses, particularly fear, anger, or moral outrage) (Brady et al., 2020). Furthermore, digital platforms can normalize these extreme viewpoints through repeated exposure (Denemark, 2023).

Exposure to extremist content on online platforms has documented negative effects on users and society. Research demonstrates that exposure to opposing political views on social media can increase political polarization rather than promote understanding (Bail et al., 2018). Additionally, social media exposure to hate speech has been linked to increased offline hate crimes, demonstrating the real-world consequences of online extremist content (Müller & Schwarz, 2021). Platform algorithms can create radicalization pathways that gradually expose users to increasingly extreme content, particularly on video platforms like YouTube (Ribeiro et al., 2020). Social factors within online conspiracy communities can draw users deeper into extremist belief systems (Phadke et al., 2021). Furthermore, victims of online hate speech experience significant psychological consequences, including increased feelings of insecurity and vulnerability (Dreißigacker et al., 2024).

This problem requires both technical solutions and governance policies promoting transparency and healthier discourse (Phadke et al., 2021). Despite regulatory initiatives like the EU's Digital Services Act and platform transparency reports, significant knowledge gaps remain due to these systems' proprietary nature. The presence of extremist narratives in discussions like BLM underscores the need for methods to predict user engagement with such content to identify at-risk individuals and develop mitigation strategies.

1.1.3 Network Structures and Echo Chambers

"Echo chambers" or "filter bubbles" in social networks are crucial for understanding online extremism (Pariser, 2011). On platforms like X/Twitter,

users engage primarily with like-minded content. Research shows conservatives, especially those with extreme views, exhibit greater ideological homophily than liberals on X/Twitter. This clustering creates environments conducive to the spread and mobilization of right-wing or extremist ideologies, thereby reinforcing echo chambers and heightening the risk of radicalization (Boutyline & Willer, 2016). Isolation reinforces radicalization through homogeneous information, group polarization, social identity reinforcement, and limited exposure to moderate perspectives.

1.1.4 Semantic Analysis of Extremist Content

Understanding the network structures and echo chambers that facilitate radicalization is crucial. However, a complete grasp of online extremism also demands an analysis of the language and content shared. The sheer volume of discourse on platforms like X/Twitter overwhelms traditional manual analysis, necessitating a shift toward scalable, data-driven methods. This leads to an exploration of how semantic analysis, driven by advances in Natural Language Processing (NLP) and sophisticated language models, offers the tools to effectively detect, classify, and interpret the subtle nuances of extremist rhetoric.

Transformer-based language models, especially BERT (Devlin et al., 2019), have significantly advanced NLP. BERT's bidirectional training provides deeper contextual and semantic understanding compared to prior methods. Building on this, a study by Alatawi et al. (2021) compared Logistic Regression, LSTM, and BERT for extremism detection on a balanced textual dataset, with BERT achieving the best performance. Similarly, Jamil et al. (2022) demonstrated that a fine-tuned BERT model alone achieved high accuracies on Twitter data, effectively identifying both overt and implicit radical language. This underscores the power of transformer-based models in capturing the nuanced semantics inherent in extremist content.

1.1.5 Graph Neural Networks and Link Prediction in Social Media

Graph Neural Networks (GNNs) have emerged as powerful tools for modeling and analyzing relational data, particularly in social media networks where users and their interactions naturally form complex graph structures (Hamilton et al., 2018). In these settings, entities such as users and tweets can be represented as nodes, while relationships such as likes or retweets serve as edges connecting them.

In link prediction tasks, GNNs aim to infer the likelihood of unseen or future connections between nodes, such as predicting whether a user will interact with a post or follow another user. This capability has made GNNs essential in applications like friend recommendation, information diffusion modeling, and community detection across social media platforms.

GNNs leverage message passing or neighborhood aggregation mechanisms, where each node updates its representation by integrating information from its connected neighbors. This enables GNNs to capture both local and higher-order structural dependencies within a network, providing a rich understanding of social dynamics. Popular architectures such as Graph Convolutional Networks (GCN) (Kipf & Welling, 2017) and GraphSAGE (Hamilton et al., 2018) have been widely used for such tasks due to their scalability and representational power.

Moreover, recent research on GNNs has shown strong effectiveness for link prediction tasks on Twitter, consistently achieving high performance across various scenarios. Praznik et al. (2021) reported an AUC of 0.959 using the SEAL GNN framework for hashtag co-occurrence prediction, outperforming traditional similarity-based methods. Alhosseini et al. (2022) achieved 94.22% precision in classifying likes, retweets, replies, and quotes using GraphSAGE on official Twitter data. Similarly, Arazzi et al. (2023) introduced TweetGage, a GNN-based model for tweet engagement prediction that effectively leveraged graph relationships among posts. Overall, GNNs provide a robust framework for social media link prediction, enabling data-driven insights into user connectivity and interaction patterns.

1.1.6 Limits of Traditional Models

Conventional approaches to predicting engagement with extremist content have relied heavily on statistical methods and machine learning algorithms that analyze content and user features independently. These methods typically employ logistic regression and random forests. However, these traditional models suffer from significant limitations:

Table 1.1 Key Limitations of Conventional Engagement Prediction Models

LIMITATION	DESCRIPTION
Contextual Blindness	Fails to capture the relational context; treats users and content as independent entities.
Feature Engineering Constraints	Relies on manually engineered features, which may overlook subtle patterns in complex data.
Modality Isolation	Analyzes text or network structure separately, ignoring the multimodal nature of content.

Sources: Alizadeh et al., 2020; Burnap & Williams, 2016; Ferrara et al., 2016; Sadeque & Bethard, 2019

These limitations have led to prediction accuracy plateaus, underscoring the necessity for novel methodologies that can capture the complex, relational, and multimodal dynamics inherent in online extremist engagement.

1.2. Statement of the Problem

The rapid spread of extremist content on social media, especially within politically charged discussions like the Black Lives Matter (BLM) movement, poses serious societal risks. While these platforms foster activism and discourse, they also amplify extremist narratives and hate speech, which have been linked to offline hate crimes and psychological harm. Despite these dangers, efforts to predict user engagement with extremist content remain limited. Existing models often analyze content and users separately, overlooking the contextual meaning of language and the relational dynamics within online networks. This results in an incomplete picture of how extremist narratives gain traction. To address this gap, this study integrates BERT for semantic representation of extremist discourse and Graph Neural Networks (GNNs) for relational modeling. Rather than examining the mechanisms of engagement formation, this research evaluates whether GNN-based approaches can effectively predict engagement patterns within extremist social media networks.

1.3. Objectives of the Study

The primary objective of this study is to evaluate the effectiveness of Graph Neural Networks (GNNs) in predicting engagement patterns between

users and tweets containing extremist narratives within BLM-related discussions on X/Twitter. Employing a dual-method approach that integrates semantic and network analysis, the study utilizes BERT for extremist tweet classification and GNNs for link prediction. This approach enables an assessment of whether GNN-based models can accurately forecast engagement within online extremist discourse networks. The study aims to:

1. **Develop and Validate a Tweet Classification Pipeline:** Manually label a subset of tweets as Extremist or Non-extremist. These binary labels were then used to fine-tune a BERT model with three output classes (Extremist, Normal, Offensive). This supervised setup leverages human judgment to guide the model while enabling broader classification across the dataset, forming a reliable basis for later network analysis.
2. **Construct and Train a GNN Model for Link Prediction:** Represent users and labeled tweets as nodes in a social graph and apply different GNN architectures under a supervised framework to predict which user-tweet pairs are most likely to form engagement links, uncovering latent pathways of extremist content diffusion.
3. **Evaluate and Compare Performance:** Systematically evaluate and compare the performance of these GNN models using standard metrics. Conduct ablation studies to identify the strengths and limitations of each architecture, providing insights into their suitability for link prediction in this context.

1.4. Scope and Limitations of the Study

Scope. This study analyzes English-language tweets with content relating to the #BlackLivesMatter from March 26 to July 24, 2020 (approximately 60 days surrounding George Floyd's death). It is important to note that while the tweets are in English, the geographic location of the users is not limited, potentially including perspectives from around the world.

Limitations. This study's exclusive focus on Twitter data excludes other online platforms and offline activities, limiting the broader generalizability of findings. The absence of deleted or suspended accounts introduces potential survivorship bias, while API rate limits may have affected the completeness of the dataset. Network analyses are constrained by the availability and granularity of metadata. Although manual labeling was essential for establishing initial ground truth, it may introduce subjectivity and inconsistencies despite the use of structured annotation guidelines. The link prediction component reflects potential user engagement with extremist content. However, such predicted interactions should not be interpreted as indicators of ideological alignment. Users may engage with extremist content for various reasons, including critique, monitoring, or counter-speech, and

interaction alone does not equate to endorsement. To ensure responsible data governance, no personal or private data beyond what is publicly available was used. The dataset was securely stored in an access-controlled environment. Ethical approval and content safety protocols were observed to minimize researcher exposure to harmful material.

1.5 Significance of the Study

This study is significant in multiple domains, particularly in social media analysis, computational social science, and digital security. By leveraging Graph Neural Networks (GNNs) to model and predict user engagement with tweets containing extremist narratives within BLM-related discussions on X (formerly Twitter), this research offers valuable insights into the mechanisms through which extremist narratives propagate online.

Academic Significance. This study advances research in social media analysis and computational social science by integrating Graph Neural Networks (GNNs) and BERT to model user engagement with extremist narratives in Black Lives Matter (BLM) discussions. It offers a novel framework combining network structure and semantic content, with potential applications to other sociopolitical contexts, including electoral rhetoric, anti-Asian hate crime narratives, and emerging social movements.

Practical Significance. The findings support improved content moderation by identifying patterns of extremist engagement, helping platforms and policymakers intervene early while respecting free expression. The study also emphasizes ethical AI, promoting fairness and transparency in the deployment of predictive models.

Chapter 2 REVIEW OF RELATED LITERATURE

2.1 Extremist Text Classification

The detection of extremist texts requires clear criteria for distinguishing extremist from non-extremist content. Recent research demonstrates that extremist texts exhibit distinctive linguistic patterns, including specific lexical choices, syntactic structures, and rhetorical strategies (Shahin, 2024). These features serve to radicalize, mobilize, and maintain group cohesion among adherents. Understanding these patterns is essential for developing effective classification systems. The following framework presents key linguistic features grounded in empirical research for systematic text classification.

2.1.1 Linguistic Features

Emotional Intensity and Lexical Characteristics. Extremist texts often use more emotional words, especially those showing anger, fear, and urgency (Bermingham et al., 2009). This includes frequent negative terms for perceived enemies, vocabulary related to threats, dehumanizing language for out-groups, and glorifying terms for in-groups (Chen, 2018).

Negation and Opposition Language. These texts build binary worldviews using systematic negation (like "not," "never"), contrasting in-group virtues with out-group flaws, and absolute language that avoids compromise (Khokhlovskaya & Chernyavskaya, 2024). This simplifies complex ideas into clear-cut categories (Borum, 2011).

Certainty and Absolutist Language. Extremist writings show high certainty through words like "must" and "will," absolute quantifiers ("all," "never"), direct statements without softening, and temporal absolutes that present conditions as permanent (Schmid, 2013).

Binary Worldview Construction. Extremist texts consistently divide reality into opposing categories. This involves clear in-group/out-group identification, moral polarization (where in-group actions are always good), oversimplified causes (blaming problems on enemy groups), and temporal binaries that contrast a pure past/future with a corrupt present (Holbrook, 2014).

In this study, tweets were classified as extremist only when they displayed two or more established linguistic features that, when combined, indicated the promotion or reinforcement of extremist discourse. Using this multi-feature threshold strengthens the validity and reliability of the dataset by ensuring that labeled content shows substantial extremist characteristics rather than isolated linguistic elements that may coincidentally appear in ordinary, non-extremist communication.

Requiring multiple features to appear at the same time helps address the challenge of distinguishing genuine extremist messages from texts that may use similar language for non-extremist purposes. Prior research shows that extremist discourse is defined by the systematic co-occurrence of several linguistic markers, not by a single feature in isolation (Vergani & Bliuc, 2015). A single linguistic feature—such as emotional intensity or binary language—may appear in legitimate political discourse, journalism, or advocacy without indicating extremism. However, the combined presence of multiple indicators forms a more reliable signature of extremist communication (Cohen et al., 2014). Cohen et al. (2014) specifically found that using multiple linguistic markers greatly reduces false positives, whereas relying on single features alone leads to frequent misclassifications.

This multi-feature method aligns with established frameworks for identifying dangerous or extremist speech, which emphasize that harmful discourse emerges from the interplay of several rhetorical techniques rather than from any single trait (Benesch et al., 2018). Research on extremist text classification further supports this approach, showing that combining lexical, semantic, and psycholinguistic features significantly improves accuracy compared to single-feature models (Devyatkin et al., 2017). By applying a multi-feature threshold, the study reduces false positives by filtering out texts that contain ambiguous or context-dependent language. This conservative classification strategy prioritizes precision and minimizes the risk of mislabeling legitimate political expression or passionate advocacy as extremist content (Soler-Company & Wanner, 2019).

2.1.2 Advanced Computational Tools

The task of extremist text classification involves identifying and categorizing textual data that promotes or incites violence, hatred, or discrimination. Various computational approaches have been explored to tackle this challenge, ranging from traditional machine learning techniques to more advanced deep learning models. In a study focused on detecting such content, Alatawi et al. (2021) examined the performance of Logistic Regression, Long Short-Term Memory (LSTM) networks, and BERT on a balanced textual dataset. Their findings contribute to the understanding of the

relative strengths of these models in classifying extremist text, demonstrating the potential of transformer-based architectures alongside more traditional methods.

Table 2.1. Performance Comparison of Different Models on Hate Speech Detection

MODEL	PERFORMANCE METRIC			
	Precision	Recall	F1-score	AUC
Logistic Regression	0.72700	0.75341	0.73996	0.81160
LSTM	0.75054	0.74727	0.74890	0.74945
BERT	0.82117	0.76034	0.78959	0.79720

Source: Alatawi et al., 2021

Based on the results, BERT outperformed the other models in terms of Precision and F1-score, indicating its superior ability to correctly identify extremist content while maintaining balanced accuracy. Although Logistic Regression achieved the highest AUC, suggesting strong overall discriminative capability, BERT's consistent performance across multiple metrics highlights the advantage of leveraging contextualized language representations in hate speech and extremism detection tasks.

2.1.3 BERT

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019), uses deep bidirectional transformers to capture the full context of words, significantly improving performance in NLP tasks. In extremist content detection, BERT's ability to analyze subtle, emotionally charged language makes it more effective than traditional models, which often rely on surface-level features. Studies by Jamil et al. (2022) and Torregrosa et al. (2022) show that fine-tuned BERT models outperform older methods in identifying radical expressions, particularly in social media environments where language is often ambiguous or coded.

Additionally, BERT embeddings can model user behavior and predict connections based on linguistic similarity, with approaches like Sentence-BERT (Reimers & Gurevych, 2019) and Siamese networks (Oniani et al., 2023) further enhancing this capability.

A major step in explainable hate speech detection is BERT base uncased HateXplain. It is built on the HateXplain dataset (Mathew et al., 2021), the first benchmark incorporating 3-class classification, target community identification, and human rationales. This dataset addresses bias and interpretability issues in traditional models. BERT fine-tuned on HateXplain improves explainability and reduces bias while maintaining performance (Mathew et al., 2021; Subramaniam et al., 2022). Practical applications show HateXplain, integrated into social media moderation systems, improves precision and recall, capturing nuanced language often missed by conventional hate speech detection methods.

2.2 Graph Neural Networks for Link Prediction

In the proposed hybrid framework, BERT’s semantic representations serve as the linguistic foundation upon which the GNN builds relational understanding. Each user or tweet node in the graph is initialized with representations derived from BERT’s classification layer, reflecting the semantic and emotional characteristics of the text. These representations are then propagated through the GNN, enabling the model to integrate both linguistic meaning and structural relationships within the network. In essence, BERT encodes what is being communicated, while the GNN learns how these communications are interconnected. This fusion allows the system to uncover latent patterns of engagement and ideological alignment that arise not only from textual signals but also from the topology of user interactions.

While Graph Neural Networks (GNNs) excel at capturing complex structural patterns and user engagement behaviors within social networks, the connection between these computational predictions and the underlying psychological motivations remains largely exploratory. GNN-based link prediction models can accurately infer the likelihood of connections between users or entities based on graph topology and learned node representations, yet whether these predicted interactions reflect genuine psychological engagement, habitual behavior, or algorithmic influence is not always clear. Addressing this question requires interdisciplinary research that bridges computational social science and psychological theory, as highlighted by González-Bailón and Wang (2016).

In this context, psychological theories have demonstrated strong explanatory power for understanding the nature and polarity of social relationships. Notably, studies integrating balance theory, status theory, and emotional information theory have successfully applied these frameworks to the task of link sign prediction, which involves classifying relationships as positive or negative. Empirical results on real-world datasets show that such models can achieve significant accuracy and reasonable performance (Kumar et al., 2019). These findings underscore the potential of combining psychological principles with graph-based learning to reveal deeper insights into user interactions and social dynamics.

GNNs provide a robust and flexible framework for link prediction in social media networks, enabling data-driven discovery of relational patterns, community structures, and user connectivity. However, fully understanding the psychological foundations of these predicted links remains an open challenge, one that calls for continued integration of theoretical psychology and graph representation learning in future research.

2.2.1 Homophily and Edge Formation

Homophily, defined as the tendency of individuals to associate with others who share similar attributes, represents a fundamental principle underlying edge formation in complex networks (Châtel et al., 2025; Apollonio et al., 2022). Within social graphs, this phenomenon implies that users exhibiting comparable interests, demographic traits, or behavioral patterns are more likely to establish connections.

Graph-based learning models operationalize this principle through node embeddings, which encode multidimensional representations of node characteristics. The degree of similarity between embeddings serves as an indicator of potential link formation, aligning with empirical findings that underscore homophily's predictive role in social tie formation (Evtushenko, 2021).

Moreover, homophily frequently coexists with the principle of triadic closure, wherein the existence of mutual connections increases the likelihood of a new edge forming, thereby reinforcing local clustering and network cohesion (Asikainen et al., 2020). Collectively, embedding similarity driven by homophily provides a theoretical and computational basis for understanding the mechanisms that govern link emergence in social networks.

2.2.2 Bipartite Homogeneous Graphs

Social media networks can often be represented as bipartite graphs, where two distinct node types, such as users and tweets, form relationships exclusively across groups rather than within them. For example, in a user–tweet graph, edges represent actions like posting, replying, or retweeting, connecting users to tweets but never users to users or tweets to tweets.

Although the underlying data consists of heterogeneous entities, the graph can be represented as homogeneous by treating all nodes as belonging to a single unified set. In this representation, both users and tweets are abstracted as nodes of the same type, and all connections, regardless of their semantic meaning, are treated as a single edge class. This approach maintains the inherent bipartite structure (edges primarily between user and tweet nodes) but allows the graph to be processed by standard homogeneous GNN architectures without the need for specialized heterogeneous layers. Conceptually, the adjacency matrix of such a graph takes the form shown in Equation 2-1.

$$A = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix} \quad (\text{Equation 2-1})$$

where:

- A is the full adjacency matrix of the bipartite homogeneous graph
- $B \in \mathbb{R}^{|U| \times |T|}$ is the incidence matrix, where $B_{ij} = 1$ if a user u_i is connected to a tweet t_j , and 0 otherwise
- B^T is the transpose of B , representing the inverse tweet-to-user relationships
- n is the dimensionality of the node embeddings
- 0 is the upper-left and lower-right blocks representing zero matrices, indicating the absence of user-user or tweet-tweet connections in the underlying bipartite structure.

In practice, even though the model treats all nodes uniformly, this bipartite pattern remains embedded within the graph topology. Node features

may still encode type-specific information, for example, user profile embeddings or tweet text embeddings, optionally augmented with a binary or one-hot node-type indicator to help the model distinguish between users and tweets during message passing.

As illustrated in Figure 2.1, this bipartite homogeneous representation enables conventional GNNs to process complex social graphs efficiently. The design simplifies computation, reduces model complexity, and ensures compatibility with common message-passing frameworks, while still preserving the cross-type relationships central to social media interactions.

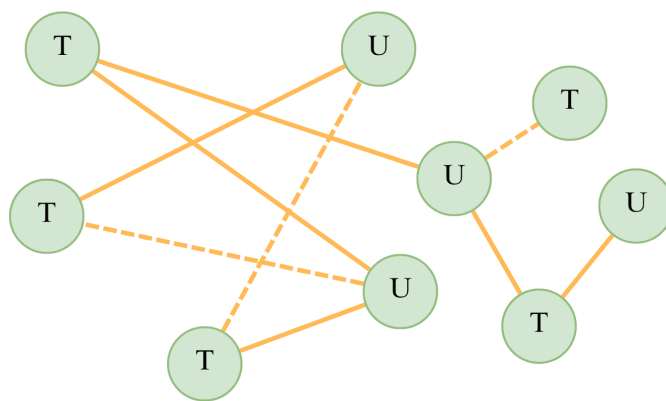


Figure 2.1. Representation of a Bipartite Homogeneous Graph with User and Tweet Nodes

Bipartite homogeneous GNNs have shown considerable promise in large-scale predictive tasks. Zhang et al. (2023) demonstrated that their DoubleH bipartite GNN achieved state-of-the-art accuracy for user stance detection on a large-scale 2020 U.S. election dataset. Similarly, Li et al. (2020) successfully deployed their HiGNN model on the Taobao platform, encompassing hundreds of millions of users and items, achieving substantial gains in e-commerce recommendation performance.

Overall, the bipartite homogeneous graph formulation provides a pragmatic balance between structural realism and computational efficiency, making it an effective strategy for large-scale social media link prediction and recommendation tasks.

2.3 GNN Architectures for Link Prediction

2.3.1 Graph Convolutional Network

Graph Convolutional Networks (GCNs) are among the earliest and most widely used GNN architectures for link prediction tasks. Introduced by Kipf and Welling (2017), GCNs operate by aggregating information from a node's local neighborhood through a convolutional framework. In social networks, this allows the model to effectively capture local structural patterns and shared node features, such as user interests or interaction histories, that are indicative of potential links. GCNs are particularly well-suited for static social graphs, where the network structure does not change frequently. Their efficiency and simplicity make them ideal for predicting friend recommendations, community detection, and collaborative filtering in well-defined and relatively stable networks.

2.3.2 Graph Attention Network

Graph Attention Networks (GATs) extend the capabilities of GCNs by introducing attention mechanisms that assign varying levels of importance to different neighboring nodes during the aggregation process. Proposed by Veličković et al. (2018), GATs learn to prioritize the most relevant connections when predicting the likelihood of a link. This is particularly valuable in social networks where not all neighbors contribute equally to a user's relationship formation, such as prioritizing close friends over acquaintances. By focusing on influential nodes, GATs improve predictive accuracy in heterogeneous and noisy social graphs, where contextual relevance is key to understanding complex link dynamics.

2.3.3 GraphSAGE

GraphSAGE, developed by Hamilton et al. (2018), introduces a sampling-based approach to neighborhood aggregation, allowing for inductive learning on large graphs. Unlike GCNs and GATs, which require a full view of the graph during training, GraphSAGE enables generalization to unseen nodes by learning how to sample and aggregate information from a fixed-size neighborhood. This makes it highly scalable and suitable for dynamic social networks with frequent updates and massive user bases. GraphSAGE has shown strong performance in tasks such as link prediction in real-time social platforms, where new users or connections are constantly introduced. Its inductive learning capability and neighborhood sampling

techniques suit dynamic and large-scale social networks (Hamilton et al., 2018; Ying et al., 2018; Wu et al., 2020). Moreover, researchers have utilized GraphSAGE in friend recommendation systems by generating node embeddings that accurately capture latent social patterns (Wang, 2022).

2.3.4 GraphSAINT

GraphSAINT, proposed by Zeng et al. (2020), is a sampling-based inductive learning method that enables scalable GNN training on large graphs. Unlike GraphSAGE, which samples neighbors layer by layer, GraphSAINT constructs minibatches by sampling complete subgraphs, allowing full message passing while avoiding neighborhood explosion. Its normalization and variance-reduction strategies further stabilize training. In link prediction tasks, GraphSAINT is particularly effective because subgraph sampling captures localized structural patterns around node pairs, improving accuracy while keeping computation efficient. Empirical results on large benchmarks such as Reddit show that it achieves state-of-the-art performance with reduced training cost, making it well-suited for dynamic and large-scale social networks (Zeng et al., 2020).

2.3.5 Comparison of GNN Architectures

Table 2.2 presents the benefits of employing different GNN architectures for link prediction within the scope of this study.

Table 2.2. Comparison of Different GNN Architectures for Link Prediction

YEAR	ARCHITECTURE	BENEFITS
2017	Graph Convolutional Network	Captures local neighborhood information efficiently, effective for static social graphs.
2018	Graph Attention Network	Uses attention mechanisms to assign different weights to neighbors when predicting links.
2018	GraphSAGE	Enables inductive learning on large graphs using neighborhood sampling.
2020	GraphSAINT	Uses subgraph sampling for minibatch

training, capturing structural patterns that enhance link prediction on large graphs.

Sources: Kipf and Welling (2017); Veličković et al. (2018); Hamilton et al. (2018); Zeng et al. (2020)

This study will explore both the Graph Convolutional Network (GCN) and GraphSAGE for link prediction. GCN efficiently captures neighborhood information through spectral-based convolutions, while GraphSAGE introduces inductive learning, making it suitable for larger or evolving graphs. By applying both architectures to the #BlackLivesMatter network, this study aims to compare their performance and determine which offers better representational capacity and predictive accuracy for this context.

2.4 Traditional Methods

Conventional approaches to predicting engagement with extremist content have relied heavily on statistical methods and machine learning algorithms that analyze content and user features independently. These methods typically employ logistic regression, random forests, or neural networks trained on features such as text characteristics, user demographics, and historical engagement patterns.

Traditional models often analyze user behavior and content separately, neglecting the interplay between them. This separation overlooks how user interactions and content context influence engagement. For instance, Sadeque and Bethard (2019) highlight that many models do not consider the relational dynamics within online networks, leading to an incomplete understanding of user engagement patterns.

Furthermore, Burnap and Williams (2016) show that while machine learning models can detect cyber hate using content features, their performance weakens when applied across diverse hate categories, suggesting the need for models that understand context and relational patterns. Ferrara et al. (2016) also observe that predicting online extremism and content adoption is limited by treating user behavior and content independently, without capturing the underlying network structures. Alizadeh et al. (2020) further reinforce this by showing that content-only models fail to fully detect coordinated influence operations, underlining the necessity of integrating semantic features with social network dynamics.

Chapter 3

METHODOLOGY

This study employs a multi-stage methodology to predict links within a social network graph constructed from X (formerly Twitter) data. The process encompasses data acquisition, preprocessing and annotation, graph construction, and the training and evaluation of a Graph Neural Network (GNN). The entire pipeline, from data collection to model training, is systematically tracked using MLflow for reproducibility.

To effectively analyze extremist content on Twitter, the study integrates two complementary approaches, as shown in Table 3.1. BERT for understanding the semantic content of tweets and classifying them into extremist (0), normal (1), or offensive (2), and GraphSAGE for analyzing the relational structure among users and tweets. This dual framework allows the model to capture both linguistic signals and the network dynamics that often characterize the spread of extremist discourse.

Table 3.1. Role of BERT and GNN

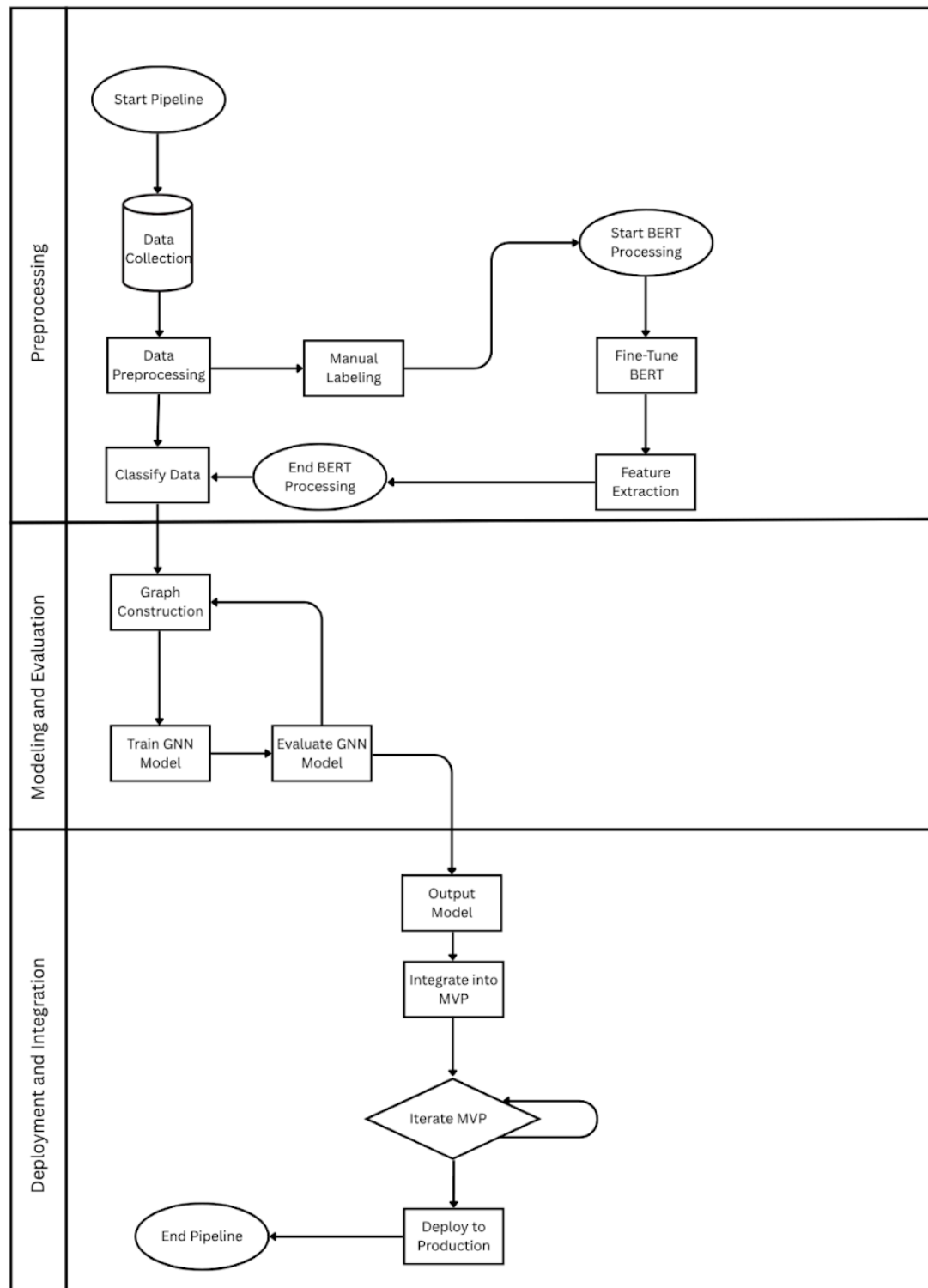
COMPONENT	ROLE	INPUT	OUTPUT
BERT (Supervised)	Classify tweets: (0) Extremist (1) Normal (2) Offensive	Tweet text	Tweet labels
GNN (Semi-Supervised)	Predict user-tweet engagement	Graph and node features	Predicted links: probability of user-tweet engagement

The HateXplain BERT model produces three predefined labels: Extremist, Offensive, and Normal. Extremist tweets explicitly promote or reinforce extremist ideology. Offensive tweets contain hostile or abusive language but do not necessarily indicate extremist intent. Normal tweets are neutral and do not exhibit offensive or extremist characteristics. These distinctions allow the model to differentiate between varying levels of harmful or neutral content.

For graph neural network (GNN) training, the dataset was consolidated into two classes to align with the manual annotation scheme. Both Offensive and Normal tweets were grouped under Non-Extremist, while only tweets meeting the criteria for extremist discourse retained the Extremist label. This approach ensures that the GNN focuses on distinguishing extremist content from all other types of tweets, improving classification precision and reducing the risk of mislabeling, while remaining consistent with the study's binary annotation framework.

The end-to-end workflow comprises three main stages: preprocessing, modeling and evaluation, and deployment and integration. Data undergoes collection, labeling, and BERT-based feature extraction before graph construction and GNN training. The finalized model is integrated into the MVP, iteratively refined, and deployed to production. This process is illustrated in Figure 3.1.

Figure 3.1. Tweet Classification and GNN-Based Link Prediction Pipeline



3.1 Data Acquisition and Storage

The graph analysis focuses on network structures containing the #BlackLivesMatter hashtag. While X's Temporal Graph Network (Rossi et al., 2020) could represent dynamic evolution over time, the current study employs static graph analysis to maintain a manageable research scope, as illustrated in Figure 3.2. The empirical basis for this study is a dataset of social media posts collected from the X platform.

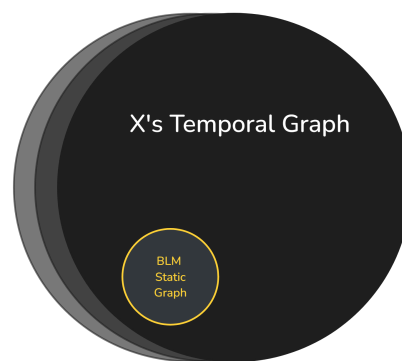


Figure 3.2. The static BLM graph is a subset of X's temporal/dynamic Graph at a certain point in time.

Data Scraping. Data collection was performed using a combination of custom scripts and established libraries. The initial data corpus was collected from English-language tweets posted between March 26, 2020, and July 24, 2020, which were filtered to include content with the #blm hashtag or replies to such content. The histories of users identified in this initial set were then systematically retrieved. The data were collected using two custom scraping modules:

- **RapidApiScraper:** Utilizes the RapidAPI interface to gather both tweet and user-level data, mainly used to build the dataset.
- **TweetyScraper:** Focuses on retrieving user metadata, specifically during inference, to enrich user node features.

Data Validation and Storage. To ensure data integrity and a centralized design, all collected records were validated against predefined Pydantic models (Tweet, User) that enforced a strict data schema. The validated data was then ingested into a self-hosted PocketBase instance, which serves as the primary data warehouse. The resulting dataset contains **36,365** records, from which the annotation process will draw its sample.

3.2 Exploratory Data Analysis

This section presents an exploratory analysis of the dataset, focusing on user-level, tweet-level, and general attributes to uncover underlying patterns, distributions, and relationships within the data.

3.2.1 User Data

The distributions in Figure 3.3 show a highly skewed pattern across most user-related attributes. The friends, favourites, following, followers, and listed counts all exhibit right-skewed distributions on a logarithmic scale, indicating that while most users have relatively low counts, a small number of users show extremely high values. This suggests the presence of highly active or influential accounts within the dataset.

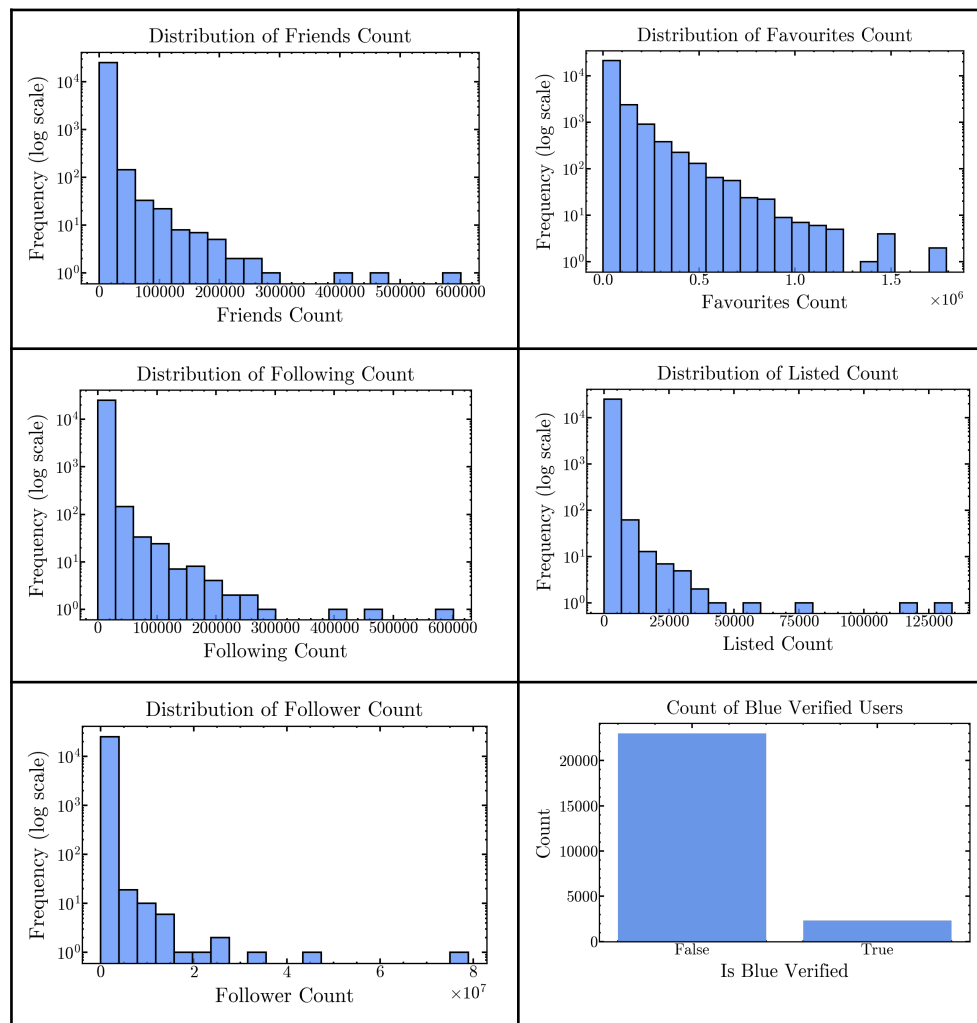


Figure 3.3. Distribution and Count of User-Level Features

The Blue Verified Status distribution reveals that only a small fraction of users hold verification, as reflected by the overwhelming dominance of non-verified accounts. The count of blue verified users further confirms this imbalance, showing that verified accounts make up only a minor proportion of the total user base.

Overall, these results show that user engagement and visibility on the platform are unevenly distributed. Most users have limited activity and reach, while a small number of highly active or well-known accounts contribute a disproportionately large share of the interactions and network activity. This imbalance highlights how influence on the platform is concentrated among a few prominent users rather than being spread evenly across the entire user base.

3.2.2 Tweet Data

The distributions in Figure 3.4 of tweet-related attributes, such as quote, reply, bookmark, retweet, and favorite counts, show a strong right-skewed pattern, indicating that most tweets receive minimal engagement while a few accumulate disproportionately high interaction levels. This highlights the unequal spread of attention typical of social media platforms, where viral or popular posts dominate overall visibility.

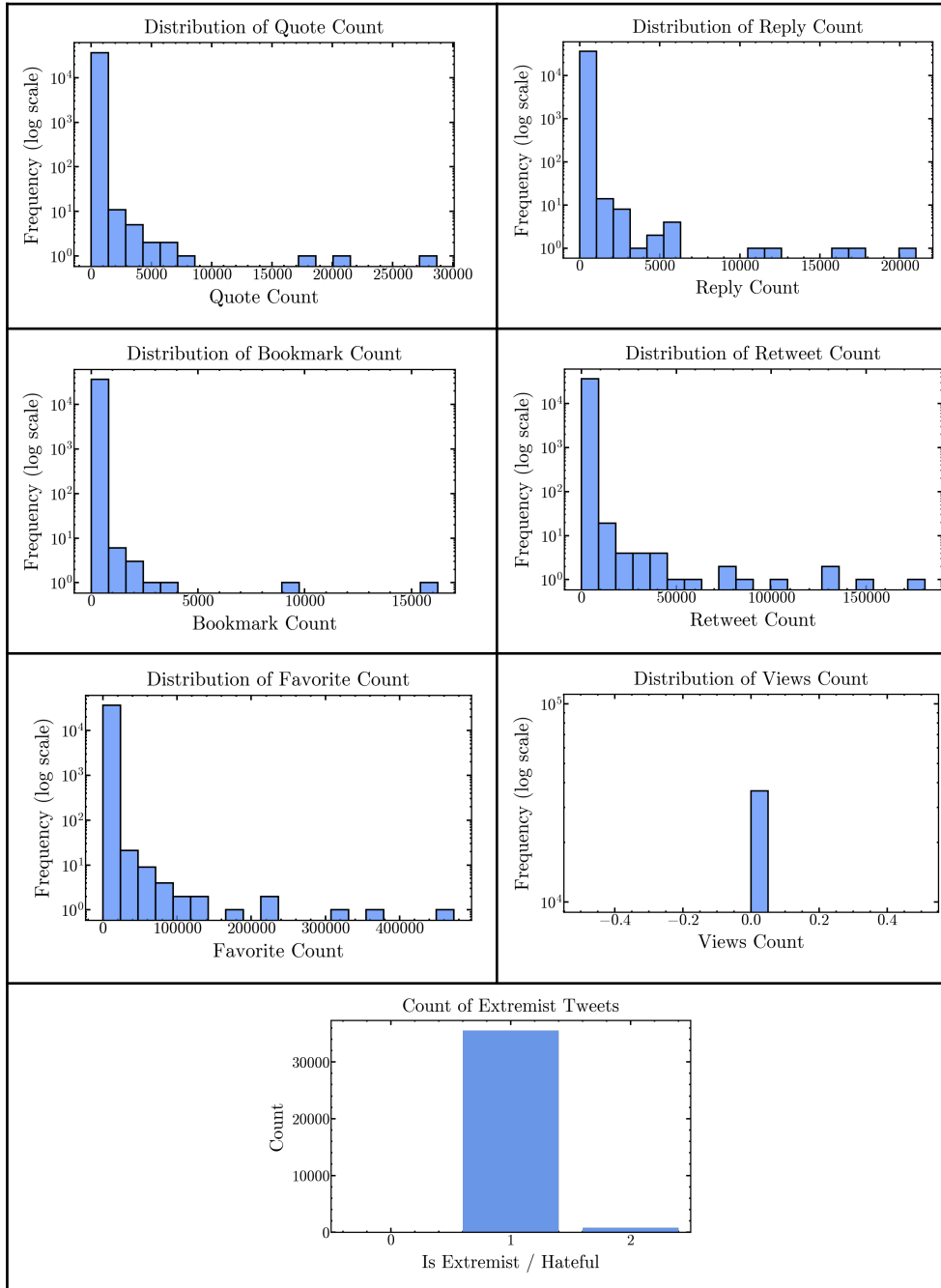


Figure 3.4. Distribution and Count of Tweet-Level Features

The views count distribution appears relatively concentrated, suggesting that most tweets have comparable or low visibility. Additionally, the distribution of extremist tweets reveals a substantial class imbalance, with extremist or hateful tweets forming only a small portion of the dataset.

Overall, the tweet-level data suggest that engagement is unevenly distributed across posts, and extremist content is relatively rare compared to non-extremist tweets. This imbalance may influence model learning and classification performance.

As shown in Table 3.2, two sample tweets illustrate how feature values can differ substantially between posts. Text represents the tweet's content; retweet count indicates how many times a tweet was shared; favorite count shows the number of likes; reply count represents user responses; quote count captures the number of retweets with commentary; and bookmark count reflects how many users saved the tweet for later viewing.

Table 3.2. Example of Tweet Data

FEATURES	TWEET 1	TWEET 2
Text	Just like we drew it up. [Image attached]	#NeverAgain was a promise broken after the #Holocaust Genocide, prejudice ...
Retweet Count	91,851	71
Favorite Count	521,169	252
Reply Count	67,554	6
Quote Count	124,743	4
Views	250,386,155	1,109
Bookmark Count	41908	0

3.2.3 General Data

The correlation matrix in Figure 3.5 shows strong positive relationships among several engagement metrics. Favorite count and retweet count have the highest correlation of 0.94, suggesting that tweets receiving more retweets also tend to gain more likes. Bookmark and quote counts also correlate moderately with other engagement metrics, while reply count shows weaker correlations overall. These patterns indicate that user engagement behaviors, such as liking, retweeting, and bookmarking, often co-occur, which reflects tweet popularity and visibility.

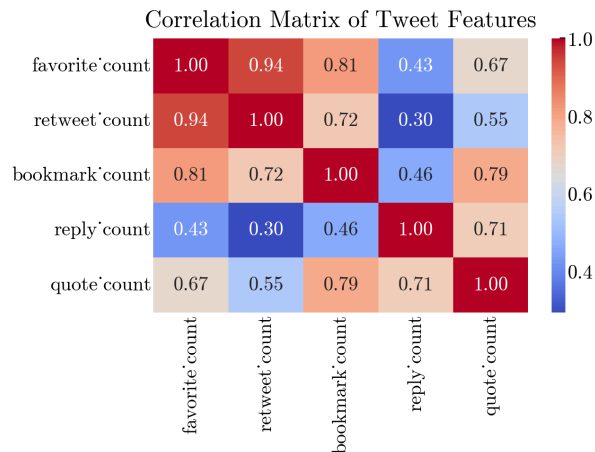


Figure 3.5. Correlation Matrix of Tweet Features

The count of tweet sources in Figure 3.6 indicates that the majority of tweets originate from unknown or mobile-based platforms, particularly Twitter for iPhone and Twitter Web App. Smaller portions come from Twitter for Android, Instagram, and third-party tools such as Hootsuite and TweetDeck. This suggests that most activity is user-generated through mobile or standard web interfaces, reflecting common patterns in general Twitter usage.

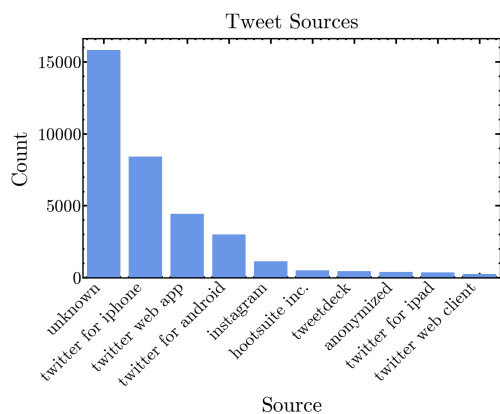


Figure 3.6. Count of Tweet Sources

The user feature correlation matrix in Figure 3.7 reveals generally weak relationships among most user attributes, except for a perfect correlation (1.0)

between following count and friends, a moderate positive correlation (0.71) between follower count and listed count, and a moderate correlation (0.50) between number of tweets and favourites count. These relationships suggest that users with larger follower bases are more likely to be included in public lists, reflecting a degree of visibility or influence. Meanwhile, the low correlations among other features indicate varied and largely independent patterns of user activity.

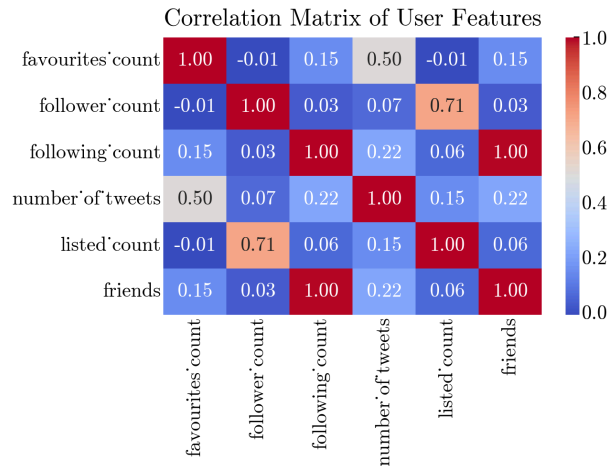


Figure 3.7. Correlation Matrix of User Features

As shown in Figure 3.8, tweets classified as Class 1 (Normal) exhibit higher average favorite and retweet counts, suggesting that these tweets are generally more popular or positively received by users. In contrast, Class 0 (Extremist) tweets receive fewer likes and shares but display higher reply activity, indicating greater user interaction through discussion or disagreement. Similar patterns are observed for quote and bookmark counts, as presented in Appendix A.

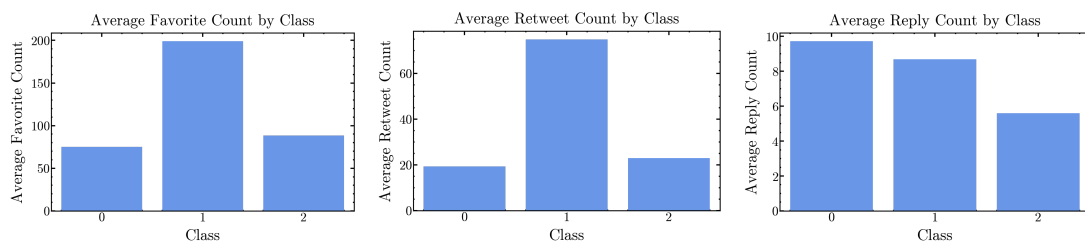


Figure 3.8. Average Engagement Metrics by Tweet Class (Favorite, Reply, and Retweet Counts)

3.2.4 Exploratory Data Analysis Insights

Overall, the exploratory data analysis reveals that both user-level and tweet-level attributes exhibit highly skewed distributions, where a small number of users and posts account for the majority of activity and engagement. This pattern suggests the presence of influential accounts and viral content that drive interactions within the network. The dominance of non-verified users underscores an uneven distribution of visibility and influence, while the limited proportion of extremist tweets indicates a pronounced class imbalance that could affect model learning and predictive performance.

Correlation analyses further show strong positive relationships among engagement metrics such as favorites and retweets, reflecting co-occurring user behaviors associated with tweet popularity. In contrast, the generally weak correlations among user features highlight the diverse participation patterns across the network. The predominance of mobile and web-based tweet sources also suggests that most interactions arise organically from typical user activity rather than automated or coordinated behavior.

In terms of engagement by class, users tend to generate more replies to tweets in Class 0 (Extremist), which may reflect disagreement or debate, while tweets in Class 1 (Normal) receive higher levels of favorites, retweets, quotes, and bookmarks, suggesting broader approval or resonance among users. Tweets categorized as Class 2 (Offensive) show relatively neutral engagement patterns, with no notable trends.

In summary, the EDA reveals an imbalanced social network structure characterized by a few highly active users and posts dominating interactions. This imbalance is an important consideration for subsequent modeling and evaluation of link prediction performance.

3.3 Data Preprocessing

The preprocessing pipeline consisted of four sequential stages aimed at preparing the dataset for downstream model training. These stages ensured that the raw Twitter data was accurately labeled, consistently structured, and transformed into graph-compatible representations. Most preprocessing tasks were handled through Pydantic models, which automated type validation and

data conversion. The only manual preprocessing involved label encoding tweet sources such as Twitter for iPhone, Twitter for Android, and Unknown for tweets without an identified source.

3.3.1 Annotation Protocol

As an initial step, a subset of tweets was manually annotated into two categories based on their textual content: Extremist (0) and Non-Extremist (1). A total of 1,700 samples, representing approximately 4.7% of the entire dataset, were annotated to support the fine-tuning of the BERT model for supervised classification. Manual annotation was limited to these two categories to maintain alignment with the study's primary objective: identifying content that explicitly promotes, supports, or reinforces extremist discourse. Including an Offensive category would have shifted the focus toward general hostility or abusive language, which does not necessarily indicate extremist intent. Limiting annotation to a binary scheme also addressed practical considerations for fine-tuning BERT. Given the relatively small number of extremist tweets, adding a third category would have further fragmented the sample, reducing the model's ability to learn distinctive extremist patterns. Focusing on Extremist versus Non-Extremist allowed for a more data-efficient and targeted fine-tuning process.

This binary labeling scheme was specifically designed to enhance the detection of extremist discourse, guided by empirically grounded linguistic features commonly associated with extremist narratives. Building on prior scholarly work in extremist text analysis, annotation guidelines were established to reliably distinguish extremist from non-extremist content using observable linguistic patterns, as summarized in Table 3.3.

Table 3.3. Linguistic Criteria for Extremist Tweet Annotation

LINGUISTIC FEATURE	DESCRIPTION
Emotional Intensity and Lexical Indicators	Presence of emotionally charged language, particularly anger, fear, or urgency. Includes dehumanizing terms for out-groups and glorifying in-groups.
Negation and Opposition Language	Use of negation (e.g., "not," "never") and oppositional framing to construct binary opposites and reinforce in-group vs. out-group narratives.

Certainty and Absolutist Framing	Statements with high certainty and lack of nuance, such as use of “must,” “will,” or absolute terms like “always” or “none.”
Binary Worldview Construction	Rhetorical framing of issues in black-and-white terms, including moral polarization, simplistic causes, and temporal dichotomies (e.g., past vs. present).

Sources: Bermingham et al. (2009); Chen (2018); Khokhlovskaya & Chernyavskaya (2024); Borum (2011); Schmid (2013); Holbrook (2014)

Tweets were labeled as extremist only when they demonstrated two or more of the identified linguistic features in a manner that clearly promoted or reinforced extremist discourse. This multi-feature requirement is supported by prior research, which shows that extremist communication is characterized by the systematic co-occurrence of several linguistic markers rather than the presence of any single feature on its own (Vergani & Bliuc, 2015). Tweets lacking these patterns or containing strong sentiment without hate or radical framing were otherwise not labeled extremist. Ambiguous cases remained unmarked and were reviewed further. Annotators were instructed to base decisions strictly on linguistic patterns, not political or ideological stance. Each tweet was labeled independently, with ethical safeguards focused on responsible handling of publicly available content and minimizing exposure to sensitive material.

3.3.2 Dataset Loader

The second preprocessing stage involved dataset management using a custom Dataset Loader utility. This module served as the interface between the PocketBase warehouse and the local training environment, ensuring efficient loading, caching, and parsing of data. Its primary purpose was to streamline data access while maintaining consistency across different experimental runs. The Dataset Loader was implemented as a Python class with two configurable parameters:

- **pb (PBWarehouse instance, optional):** If provided, this enabled direct retrieval of records from the PocketBase warehouse. In cases where this parameter was not specified, methods dependent on live fetching raised errors, thereby enforcing the need for proper warehouse linkage.
- **cache_dir (Path, optional):** A directory used to locally cache datasets for repeated access. By default, this directory was

located in the project's .cache folder, ensuring persistent availability of previously retrieved datasets.

The Dataset Loader streamlined the data workflow by handling both initial data fetching and subsequent access. It could load data from the local cache for rapid access or directly from the PocketBase warehouse to ensure the most up-to-date records were used. This dual capability guaranteed both reproducibility and efficiency. Furthermore, the module offered flexibility by returning the dataset in different formats, including a Polars DataFrame or a NetworkX Graph, which allowed for seamless integration into various downstream preprocessing and analysis pipelines.

3.3.3 Preprocessor

The third stage of preprocessing involved systematic cleaning and transformation of the dataset using a custom Preprocessor class. This module was designed to operate on data stored as a Polars DataFrame and ensured that raw attributes were standardized into machine-readable formats suitable for both downstream modeling and graph construction.

This stage handled the refinement of the source column, which contained metadata about the platforms used for posting tweets, by removing embedded HTML, standardizing source strings into categories such as "unknown," "anonymized," or lowercase identifiers, and encoding them into numerical values for model integration. Beyond source normalization, the Preprocessor applied broader transformations to enhance dataset consistency, such as encoding categorical variables, cleaning text to remove extraneous characters, reformatting attributes to match schema requirements, and standardizing numerical features to reduce skewness and ensure comparability. By centralizing these transformations, the Preprocessor ensured a modular, reproducible data pipeline that reduced variability in downstream model performance and improved consistency across experiments.

3.3.4 Graph Builder

The final preprocessing step involved constructing a graph representation of the dataset using the Graph Builder module. This stage was critical in transforming tabular tweet-user records into a structured graph format that captured both textual attributes and relational dynamics. By integrating content features with interaction patterns, the resulting graph provided the foundation for graph neural network training.

The Graph Builder operated on a Polars DataFrame containing user and tweet information, along with a feature specification object that defined which attributes should be assigned to each node type. The module validated input data to ensure it was properly formatted, non-empty, and free from null values before initiating graph construction. Graph construction supported multiple configurations, shown in Table 3.4, allowing flexibility in how relationships were represented.

Table 3.4. Graph Builder Configuration and Properties

CONFIGURATION	DESCRIPTION
Homogeneous vs. Heterogeneous	Graphs could be built as homogeneous (single node and edge type) or heterogeneous (multiple node and edge types).
Directed vs. Undirected	Graphs could optionally be directed, enabling explicit modeling of interaction direction (e.g., user $\rightarrow$ tweet).
Node Types	Nodes represented either users or tweets, each with associated features and class labels.
Edge Types	Edges captured interactions such as authorship or engagement, with the option to distinguish among types in heterogeneous graphs.
Feature Representation	Node attributes were extracted as tensors, with padding applied to maintain consistent dimensionality.

By integrating both semantic and structural properties, the Graph Builder generated bipartite graphs that modeled the interplay between users and their tweets. This design allowed downstream graph neural networks to learn not only from individual node features but also from the relational patterns that emerged across the social network.

3.4 BERT (HateXplain)

The study employed the HateXplain model, a BERT-based architecture specifically developed for hate speech and offensive language detection. Unlike conventional BERT models that rely solely on categorical labels during training, HateXplain incorporates human-annotated rationales. These are text segments that explicitly explain why a particular instance is classified as hate, offensive, or neutral. These rationales provide an additional layer of interpretability by highlighting the contextual cues that influenced human judgment. Examples from the HateXplain dataset are shown in Figure 3.9. Consequently, the model not only learns from labels but also from the reasoning behind those labels, leading to more explainable and reliable predictions.

Text	Dad should have told the muzrat whore to fuck off , and went in anyway
Label	Hate
Targets	Islam
Text	A nigress too dumb to fuck has a scant chance of understanding anything beyond the size of a dick
Label	Hate
Targets	Women, African
Text	Twitter is full of tween dikes who think they're superior because of "muh oppression." News flash: No one gives a shit.
Label	Offensive
Targets	Gay

Figure 3.9. Examples from the HateXplain dataset. The highlighted portion of the text represents the annotator's rationale. *Source:* Mathew et al. (2021)

HateXplain's architecture, as illustrated in Figure 3.10, retains the standard BERT encoder layers but introduces a multi-task learning framework, jointly optimizing for both label prediction and rational identification. This approach encourages the model to attend to semantically meaningful tokens rather than relying on superficial patterns or word co-occurrences. However, a known limitation of HateXplain is its lack of awareness of conversation-level context. Since it processes individual text inputs independently, it cannot fully capture nuances embedded in reply chains, dialogues, or extended user interactions, which is a significant consideration in the analysis of social media discussions such as Twitter threads.

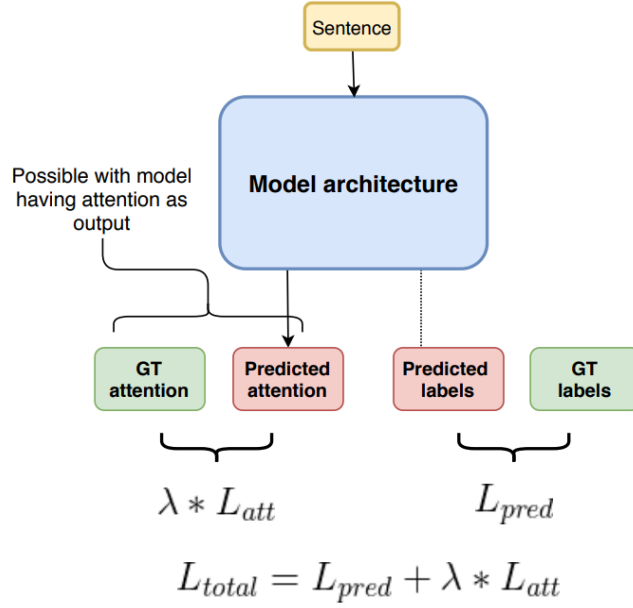


Figure 3.10. HateXplain Model Architecture. *Source:* Mathew et al. (2021)

3.4.1 BERT Experimental Setup

This study conducted a series of experiments to evaluate the effectiveness of combining BERT HateXplain embeddings with Graph Neural Networks (GNNs) for link prediction in extremist social media networks. Each configuration was tested across 50 independent trials to ensure statistical reliability and minimize the effects of random initialization. The experiments differed primarily in how the embeddings were obtained.

Experiment 1 using Pre-trained BERT. The first experiment used the pre-trained HateXplain model to extract contextual embeddings directly from the dataset. This setup served as the baseline, enabling assessment of performance without dataset-specific adaptation.

Experiment 2 using Fine-tuned BERT. In the second experiment, the HateXplain model was fine-tuned on a combined dataset consisting of manually annotated tweets and pseudo-labeled samples generated from unlabeled tweets using the base HateXplain model. This hybrid training set expanded the dataset while preserving the semantic knowledge encoded in the original model. Fine-tuning was performed over 32 epochs with a batch size of 16 and a learning rate of 5e-5, with the HateXplain classification head unfrozen. To enhance performance on the minority class (Class 0 – Extremist), oversampling increased its sample count from 53 to 809, and word-level dropout with a probability of 0.2 was applied to each token.

(whitespace-separated) 15 times per tweet. These steps aimed to improve robustness and mitigate class imbalance during training.

Comparisons between the baseline and fine-tuned results provide insight into whether the inclusion of pseudo-labeled samples improved classification accuracy and robustness, while also highlighting potential trade-offs associated with the added labeling noise.

3.5 Graph Neural Network

The study implemented two distinct Graph Neural Network (GNN) architectures: GraphSAGE and the Graph Convolutional Network (GCN), to model relational structures between users and tweets. Both models followed a similar processing pipeline, differing primarily in how they aggregate neighborhood information during embedding generation.

General GNN Architecture. In both architectures, node embeddings were initialized from preprocessed features and passed through successive graph layers, as illustrated in Figure 3.11. For the GraphSAGE model, embeddings were updated through an aggregate-and-update mechanism, where each node learned representations by sampling and aggregating feature information from its neighboring nodes. Conversely, the GCN model employed a spectral convolution approach that performed a weighted averaging of neighboring node features, enabling the network to capture local graph topology through smooth feature propagation.

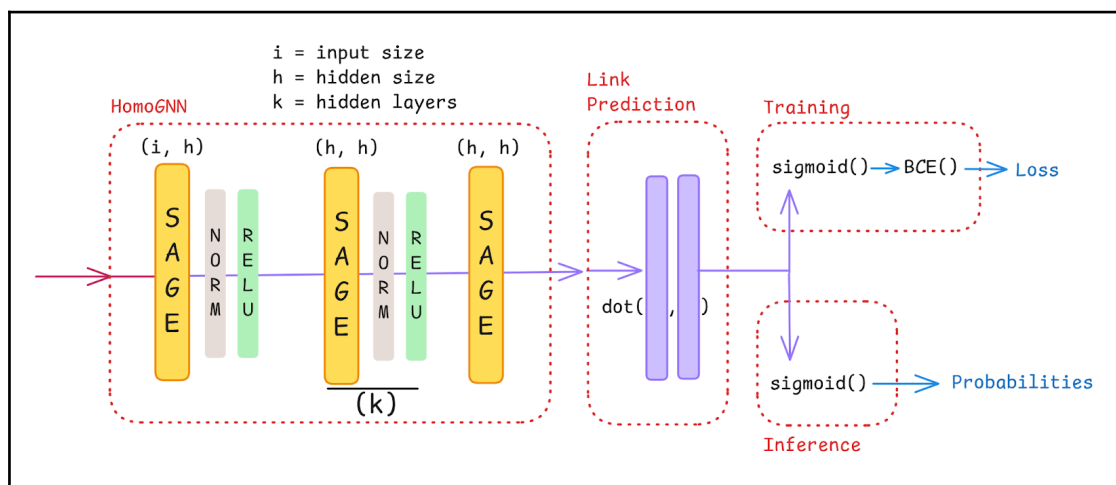


Figure 3.11. General Graph Neural Network (GNN) Architecture. The architecture is shared by both GraphSAGE and GCN implementations, with SAGE layers substituted by GCN layers when employing the GCN model.

After each graph layer, the output was batch-normalized and passed through a ReLU activation function to stabilize learning and introduce non-linearity. The output layer for both architectures excluded batch normalization and activation to preserve embedding expressiveness and ensure that the learned representations remained suitable for downstream link prediction. Since both user and tweet nodes needed to share a consistent input dimensionality, zero-padding was applied when necessary to align feature vectors. This uniform dimensional structure enabled efficient message passing and ensured model compatibility across node types.

Equation for Link Prediction. The probability of a link existing between two nodes i and j is computed in Equation 3-1 using the dot product of their learned embeddings, followed by a sigmoid activation to normalize the values into a probability range of $[0, 1]$. Unlike cosine similarity, which measures only the directional closeness of embeddings, the dot product captures both direction and magnitude, enabling a more expressive representation of structural relationships in the graph. This approach allows the GNN (whether GraphSAGE or GCN) to better learn how users and tweets interact within the network.

$$\hat{y}_{ij} = \sigma(z_i^T z_j) \quad (\text{Equation 3-1})$$

where:

$\hat{y}_{ij}$ is the predicted probability of a link between nodes i and j

$\sigma(\cdot)$ is the sigmoid function defined as $\sigma(x) = \frac{1}{1+e^{-x}}$

$z_i, z_j \in \mathbb{R}^n$ are the embedding vectors of nodes i and j , respectively, obtained from the final layer

n is the dimensionality of the node embeddings

$z_i^T z_j$ is the dot product between the embeddings, representing the raw similarity (logit) before normalization

During training, the resulting logits were passed to the binary cross-entropy (with logits loss) function to optimize model parameters by minimizing the difference between predicted and actual link labels (1 for existing links, 0 for non-existing ones). The same loss function also served as a link validation to evaluate how well the model generalized to unseen data.

This ensured consistent measurement of the model’s ability to distinguish between positive and negative links across both training and validation phases. Stochastic Gradient Descent was used as the optimizer with 0.9 momentum and $5e-4$ weight decay.

During inference, the logits were processed through a sigmoid function to convert them into interpretable probability scores, indicating the likelihood of a link existing between two nodes. Although the models could theoretically predict all possible node connections, including self-links and user-to-user interactions, the study specifically focused on user-to-tweet relationships. Therefore, a filtering mechanism was implemented during inference to retain only user-to-tweet probability scores, excluding irrelevant connections.

Significance of the GNN Architectures. The inclusion of both GraphSAGE and GCN architectures served a comparative purpose. GCN, being a transductive model, excels in learning representations when the entire graph is known during training, making it well-suited for dense, static datasets. GraphSAGE, on the other hand, is an inductive model that generalizes to unseen nodes by sampling local neighborhoods, making it more scalable and adaptable to dynamic or incomplete graphs. By evaluating both architectures under the same experimental conditions, the study aimed to identify which framework better captured relational patterns and generalization behaviors within social media interaction networks.

3.5.1 Graph Splitting

The supervised training procedure relies on dividing the dataset into three distinct splits: training, validation, and testing, with ratios of 0.85, 0.05, and 0.10, respectively. Each split contains the full set of nodes but only a disjoint subset of edges. During the training phase, 85% of the edges are used for message passing, while 5% are allocated for supervision. In the validation phase, message passing is performed using the union of the training message-passing edges and training supervision edges, and the remaining 10% of edges serve as validation supervision.

Each phase requires two types of supervision edges: positive and negative. Positive edges correspond to actual connections present in the data and are assigned a label of 1, while negative edges correspond to non-existent connections and are assigned a label of 0. This framing renders link prediction a binary classification problem. The total number of possible edges is computed by multiplying the number of source nodes by the number of potential destination nodes. For the bipartite directed graph in this study, this

is expressed as $|U| \times |T|$, where U and T represent the sets of user and tweet nodes.

As shown in Table 3.5, the dataset contains four node types with the following frequencies:

Table 3.5. Frequency Table of Node Types

NODE LABEL	NODE TYPE	COUNT
0	Extremist Tweet	54
1	Normal Tweet	35,553
2	Offensive Tweet	758
3	User	25,317

Based on these counts, the dataset contains 36,365 positive edges. Since $|T|$ is the combined number of extremist, normal, and offensive tweets, the total number of possible edges is 920,652,705. Subtracting the number of observed positive edges yields 920,616,340 negative edges. Positive edges, therefore, constitute far less than 0.00001% of all possible edges, indicating extreme sparsity in the adjacency matrix.

In implementation, adjacency information is typically stored in Coordinate (COO) format, a representation well-suited for sparse matrices. In this format, all existing edges are stored as parallel arrays containing source and destination indices. Negative edges require no explicit storage because their absence in the edge list is equivalent to being assigned a label of 0. This approach enables efficient and scalable processing for large, sparse graphs.

The distribution of incoming edges per tweet class is summarized in Table 3.6, which provides both the observed counts and their corresponding number of possible edges.

Table 3.6. Incoming Edge Counts for Each Tweet Class

NODE LABEL	NODE TYPE	INCOMING EDGES	POSSIBLE EDGES
0	Extremist Tweet	35,553	1,367,118
1	Normal Tweet	758	900,095,301
2	Offensive Tweet	54	19,190,286

Each tweet contains exactly one incoming edge, meaning that only a single user interacts with each tweet in the dataset. As a result, the model can learn only how particular user types map to tweet types, while deeper or latent network patterns remain unobservable. Increasing the dataset size or incorporating additional user-tweet interactions would help address this limitation.

3.5.2 Ablation Study

To evaluate the model’s generalizability and isolate the impact of graph architecture and hyperparameter choices, an ablation study was conducted using embeddings from the pre-trained BERT HateXplain model. By relying on pre-trained embeddings, the study avoided the confounding effects of fine-tuning and focused solely on the contribution of GNN design decisions.

The ablation study systematically varied key components, including graph orientation, GNN architecture, hidden dimensions, number of layers, aggregation methods, and classification thresholds. Each configuration was evaluated across 50 independent runs over 500 training epochs to ensure stable and reliable performance estimates.

The baseline settings for these experiments were chosen based on standard practices and prior literature in graph-based learning. The GraphSAGE and GCN architectures were selected based on foundational work by Hamilton et al. (2018) and Kipf and Welling (2017), while hyperparameter ranges were informed by comprehensive benchmarking

studies (Chen et al., 2024; Wu et al., 2024). Table 3.7 summarizes these baseline values alongside the ranges explored during the ablation study.

Table 3.7. Baseline Configuration for the Ablation Study

ASPECT	BASE VALUE	CONFIGURATION RANGE
GNN Architecture	GraphSAGE	GraphSAGE, GCN
Hidden Dimension Size	24	2-256
Hidden Layers	8	2-64
Classification Threshold	0.5	0.0-1.0
Graph Orientation	Directed	Directed, Undirected
Aggregation Method	Mean	Mean, Median, Max, Sum

3.6 Model Deployment and Visualization

After model training and inference, the next step was to make the results accessible and interpretable through deployment and visualization. This phase focused on transforming the raw outputs of the models into a functional system that users can interact with, while also presenting findings in clear and actionable visual formats.

Deployment Process. The deployment followed an iterative Minimum Viable Product (MVP) approach, with three successive versions:

1. **MVP 1.0 CLI Tool** – A command-line interface that allowed users to run inference directly and return results as tuples of PocketBase records with their corresponding probabilities. This version served as proof of concept for the inference pipeline.
2. **MVP 2.0 Streamlit Dashboard** – Introduced a basic web interface for running inference. It is connected to a PocketBase database, enabling easier interaction with results compared to the CLI tool.
3. **MVP 3.0 Production Build** – The final version, integrating both the backend inference engine and a polished frontend web

application. It included study details, author information, and user-facing functionality.

Deployment Framework and Tech Stack. The system architecture shown in Figure 3.12 illustrates the deployment framework and technology stack used for the MVP 3.0 Production Build. The application adopts a modular, decoupled design, separating the frontend, backend, and database components to enhance scalability and maintainability. The frontend, built with Svelte and hosted on Cloudflare Workers, provides a responsive and lightweight user interface. The backend, implemented with FastAPI, handles model inference and exposes REST endpoints for classification and prediction tasks. The PocketBase database manages persistent storage, including user credentials, records, and inference results. This architecture enables efficient data flow between components while supporting flexible deployment both locally and in cloud environments such as Railway.



Figure 3.12. System Architecture

Visualization of Results. Visualization played a critical role in making the model outputs interpretable. The inference results were visualized through performance metrics and network-based insights. For instance, charts showing accuracy, precision, recall, and F1-score were plotted for each label (extremist, normal, offensive) to demonstrate the classification reliability. Confusion matrices were also used to provide a clear breakdown of correct and incorrect predictions, which made the model's behavior transparent.

User Dashboard. To support accessibility, a simple and interactive dashboard was created to serve as the visualization front end. It was designed to present the model's findings clearly and intuitively for both technical and non-technical users. The dashboard showcased model performance metrics and visualizations, allowing users to better understand the results. The overall user flow is illustrated in Figure 3.13.

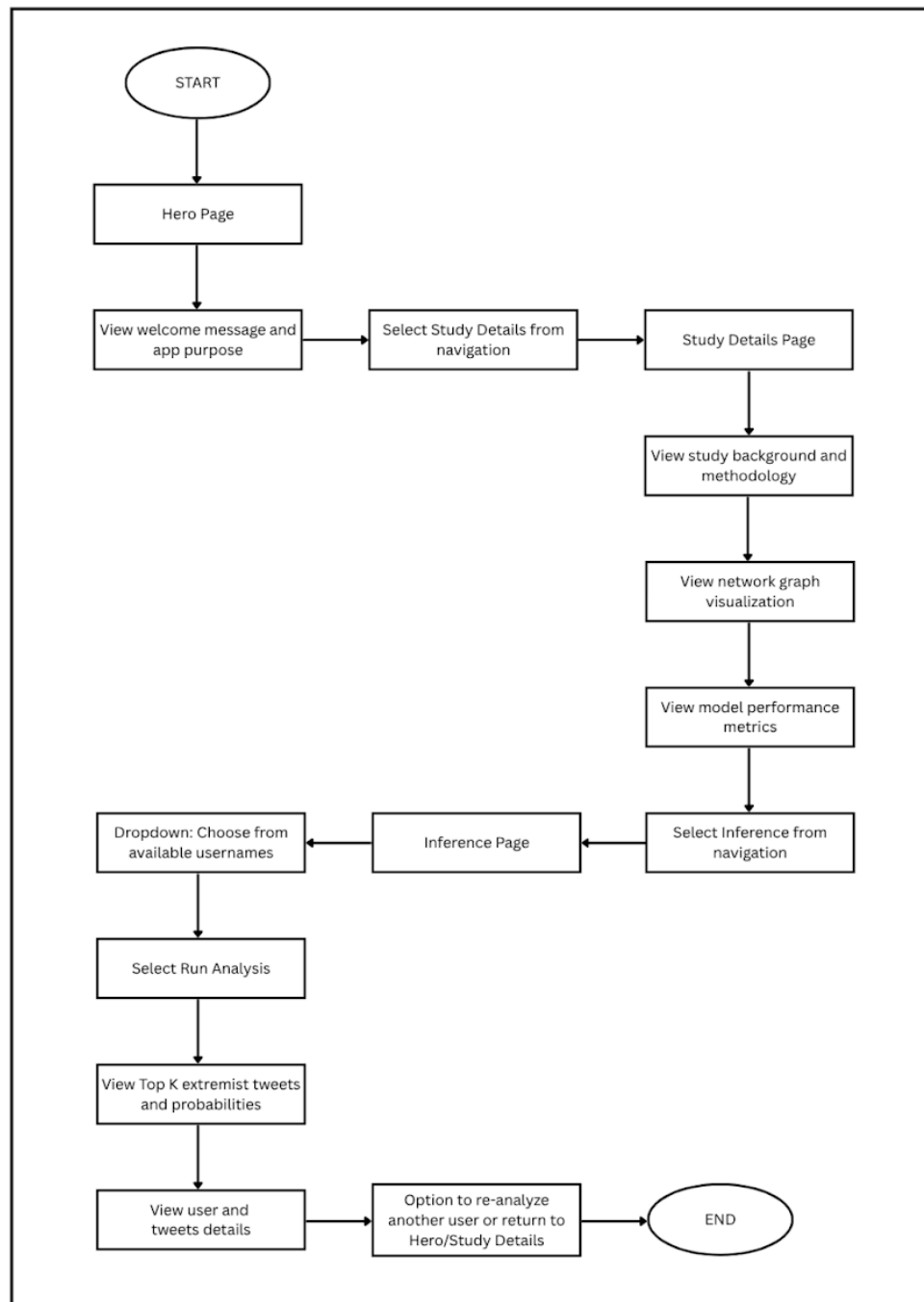


Figure 3.13. User Flow Diagram

Practical Implications. By combining deployment with visualization, the project delivered more than a theoretical model. It provided a working tool that translates machine learning results into actionable insights. Stakeholders can use the system to detect extremist discourse, monitor potential diffusion pathways, and evaluate intervention strategies in real-time. For access to the deployed prototype and dashboard, see Appendix B.

Chapter 4

RESULTS AND DISCUSSIONS

This chapter presents the results of two major experimental components: the BERT HateXplain embeddings experiment and the subsequent ablation study. The first set of experiments evaluates the effects of using pre-trained and fine-tuned BERT HateXplain embeddings, establishing how different embedding strategies influence model performance. Next, an ablation study is conducted to systematically examine the contribution of individual architectural components to the overall model design. The best-performing model from these investigations is then applied to the link prediction task in the extremist social media network, providing the final evaluation of the proposed approach.

4.1 Results of BERT Experiments

This section presents the outcomes of the BERT embedding experiments, comparing the performance of the pre-trained HateXplain model and the fine-tuned variant on the task of extremist content classification. Key evaluation metrics, including Accuracy, Precision, Recall, F1 Score, and PR-AUC, are reported and analyzed to assess the impact of domain-specific fine-tuning and the incorporation of pseudo-labeled samples.

Experiment 1. As shown in Figure 4.1, Class 0 (Extremist) achieved an F1 score of 0.686, indicating moderate performance in detecting extremist tweets. Class 1 (Normal) obtained the highest score at 0.998, reflecting near-perfect identification of normal tweets. Class 2 (Offensive) followed with an F1 score of 0.922, demonstrating strong performance in classifying offensive tweets. These results indicate that the pre-trained HateBERT model performs exceptionally well for the majority classes (Classes 1 and 2), while performance declines for the minority extremist class (Class 0), likely due to class imbalance and limited representation in the training data.

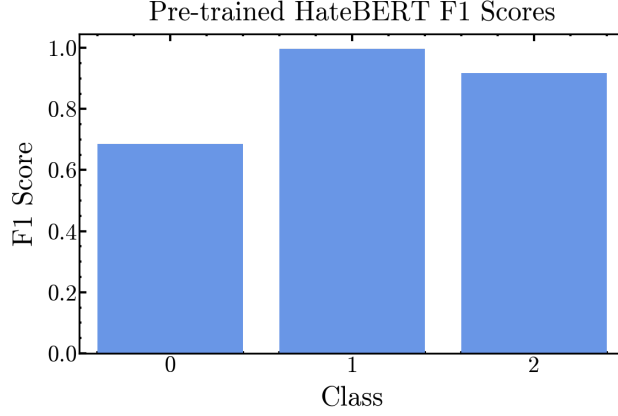


Figure 4.1. Pre-trained HateBERT Classification Results using F1 Score

Experiment 2. As shown in Figure 4.2, the fine-tuned BERT model achieves an F1 score of 0.787 for Class 0 (Extremist), surpassing the baseline pre-trained result of 0.686. Class 1 (Normal) retains strong generalization, while Class 2 (Offensive) shows a slightly lower performance with an F1 score of 0.7647. Overall, the fine-tuning process successfully improved the detection of extremist content; however, the performance boost for Class 0 may partially reflect overfitting due to the effects of oversampling.

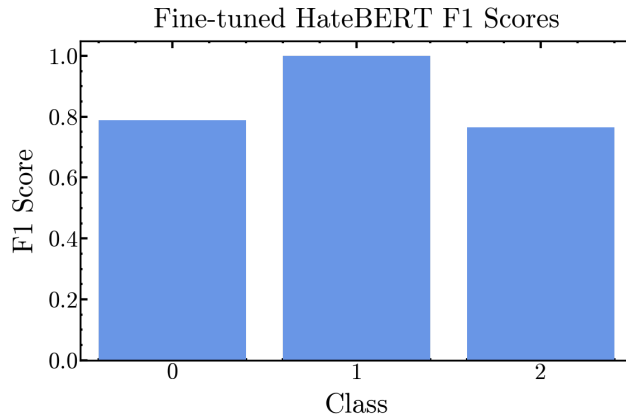


Figure 4.2. Fine-tuned HateBERT Classification Results using F1 Score

4.2 Results of Ablation Study

This section summarizes the findings from the ablation study conducted on the Graph Neural Network architectures. The study systematically evaluated the contribution of individual components, such as

hidden layer size, number of layers, aggregation method, and graph orientation, to model performance. Results are presented to identify the configuration that optimally balances accuracy, generalization, and interpretability for link prediction in extremist social media networks.

4.2.1 GNN Architecture

Figure 4.3 compares the performance of GraphSAGE and GCN across Accuracy, F1 Score, and PR-AUC over 500 training epochs.

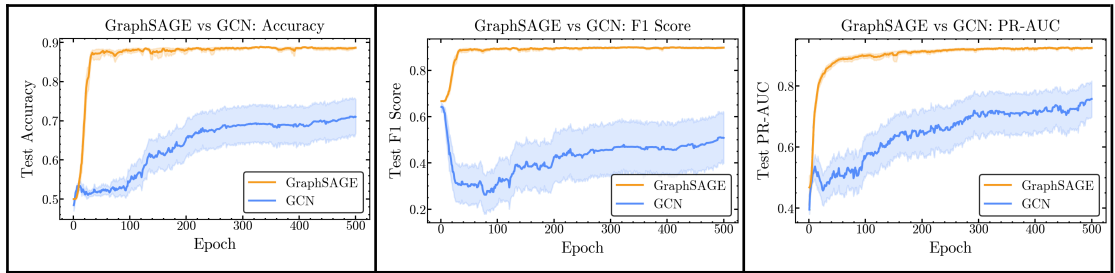


Figure 4.3. Comparison of GraphSAGE vs. GCN Architectures on Accuracy, F1-Score, and PR-AUC

In terms of Accuracy, GraphSAGE rapidly converges within the first 50 epochs, reaching a stable performance of approximately 0.88. In contrast, GCN demonstrates a slower learning curve, gradually improving from near-random initialization and stabilizing around 0.70. This indicates that GraphSAGE is more effective in capturing node-level features that contribute to accurate classification.

The trend is consistent in the F1 Score plot. GraphSAGE quickly achieves a near-optimal F1 score of approximately 0.90, suggesting that it balances precision and recall effectively. GCN, however, lags behind, plateauing at approximately 0.5 after 500 epochs with higher variance. This gap highlights GraphSAGE's stronger capability in handling imbalanced classes or noisy neighborhoods, where capturing discriminative node representations is crucial.

A similar observation is evident in PR-AUC. GraphSAGE reaches a high PR-AUC value above 0.9 early in training and maintains it consistently, whereas GCN only improves gradually to approximately 0.7 with larger fluctuations. Since PR-AUC is particularly sensitive to false positives, the

results suggest that GraphSAGE is more reliable in ranking positive edges higher than negatives, making it advantageous in link prediction or edge classification tasks.

Table 4.1 presents the model performance using pre-trained BERT HateXplain labels as node features for the link prediction task. Between the two graph neural network architectures evaluated, a substantial performance gap is observed across all evaluation metrics.

Table 4.1. Model Performance of GCN and GraphSAGE using Pre-trained BERT HateXplain Embeddings

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	PR-AUC
GCN	0.710	0.471	0.567	0.508	0.758
GraphSAGE	0.887	0.815	0.999	0.898	0.925

The GraphSAGE model achieved the highest overall performance, with an accuracy of 0.887, F1-score of 0.898, and PR-AUC of 0.925, indicating strong capability in correctly identifying both positive and negative links within the graph. Its recall score of 0.999 suggests that nearly all true positive links were successfully detected, though the slightly lower precision (0.815) indicates a modest number of false positives. This behavior implies that GraphSAGE effectively generalizes the contextual and relational features learned from the BERT HateXplain embeddings, capturing fine-grained semantic cues and user-tweet relationships within the network.

In contrast, the GCN model exhibited considerably lower performance across all metrics, with an accuracy of 0.710 and an F1-score of 0.508. This result suggests that GCN struggled to effectively propagate node information through the network. Unlike GraphSAGE, which aggregates neighbor features in a more adaptive and inductive manner, GCN relies on fixed graph convolutional operations that may oversmooth node representations, leading to weaker differentiation between extremist and non-extremist connections.

Overall, these findings demonstrate that GraphSAGE not only delivers superior predictive performance but also provides more consistent and

trustworthy results. This makes GraphSAGE the more effective architecture for the task, particularly in scenarios where both accuracy and stability are critical.

4.2.2 Hidden Dimension Size and Layers

Figure 4.4 illustrates the effect of hidden dimension size on model performance. For Test PR-AUC, all larger dimensions (8–256) converged to similar values (~ 0.90 – 0.95). Only the very narrow configurations (2 and 4) consistently underperformed, failing to surpass 0.85. This suggests that PR-AUC is largely insensitive to dimension size once a sufficient capacity is reached.

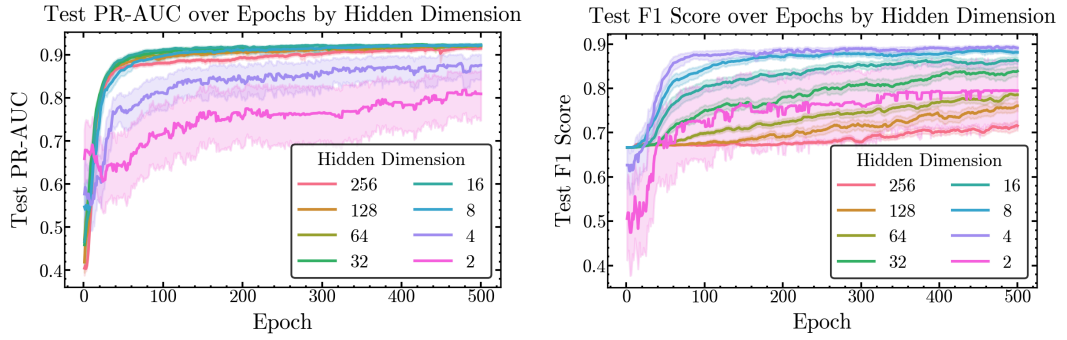


Figure 4.4. Impact of Hidden Dimension Size on Test PR-AUC and F1 Score Over Training Epochs

By contrast, Test F1 Score reveals a more nuanced trend. The model with a hidden size of 4 achieved the highest F1 Score of approximately 0.89, followed closely by 8 (≈ 0.87 – 0.88) and 16 (≈ 0.85 – 0.86). Models with larger hidden dimensions (32 and above) consistently underperformed, plateauing around 0.83–0.84 and exhibiting higher variance. These results indicate that while higher dimensions do not harm PR-AUC, they may introduce inefficiencies or overfitting that reduce balanced performance. Consequently, a smaller hidden size (8 or 16) appears optimal, striking the best balance between precision and recall for this dataset.

Figure 4.5 shows the analysis of model depth, which revealed an inverse relationship between the number of hidden layers and predictive performance. The models were trained with 2, 8, 16, and 32 layers using a learning rate of 0.1. However, when extended to 64 layers, the model became numerically unstable, producing NaNs in the output. To address this, the

learning rate was reduced to 0.01, which stabilized training but resulted in slower convergence and lower predictive performance relative to the shallower configurations.

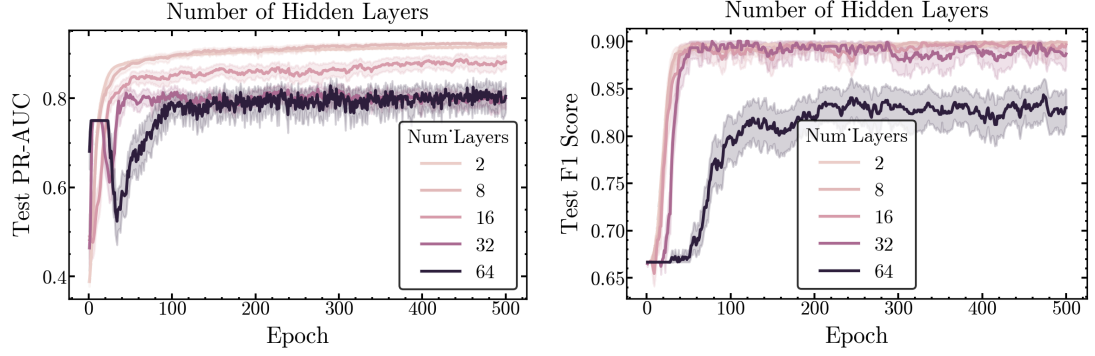


Figure 4.5. Effect of Number of Hidden Layers on Test PR-AUC and F1 Score Over Training Epochs

As depth increased, performance declined consistently, with the 256-layer model yielding the lowest average F1 score of approximately 0.65 and exhibiting the greatest variance. This trend indicates that deeper graph neural networks are susceptible to issues such as over-smoothing or overfitting on this dataset, suggesting that optimal model complexity lies in shallower architectures.

Figure 4.6 illustrates the impact of hidden dimension size and number of layers on training run duration and VRAM usage. Both metrics increase almost linearly with model complexity. The training time exceeds 400 seconds at 256 hidden dimensions and 300 seconds at 64 layers. Increasing the hidden dimension results in a sharp growth in memory requirements, surpassing 3000 MB at the maximum configuration. The number of layers also drives up VRAM usage linearly, reaching about 2000 MB at 64 layers. Hidden dimensions exert a stronger influence on memory consumption compared to layer depth, reflecting their greater contribution to parameter size.

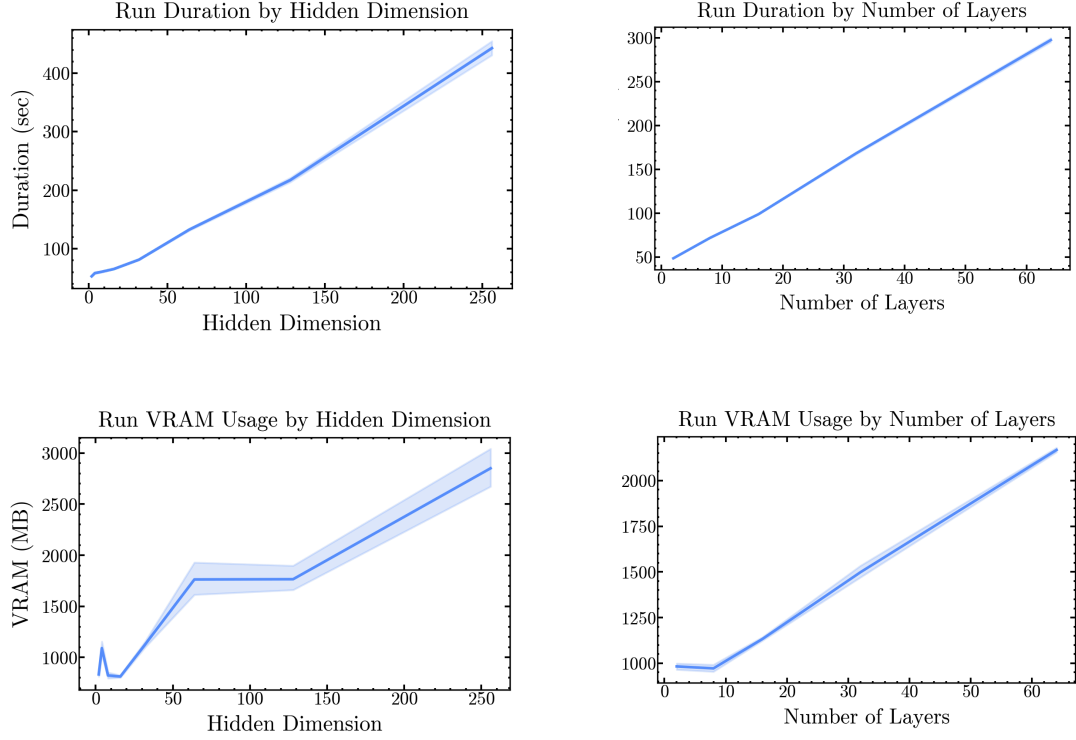


Figure 4.6. Impact of Hidden Dimension Size and Number of Layers on Training Duration and VRAM Usage

While these results confirm predictable scaling behavior on the tested dataset, it remains uncertain how the model would perform on larger datasets, such as the full X/Twitter graph. Nonetheless, the observed linear relationship provides a practical basis for balancing model complexity with available computational resources.

4.2.3 Classification Threshold

Figure 4.7 illustrates how varying the classification threshold affects model performance for both directed and undirected graph representations. The directed model reaches its highest F1 Score—exceeding 0.8—within the 0.5–0.7 range, peaking near 0.7. In contrast, the undirected model shows more stable performance across thresholds but never surpasses the directed model’s peak.

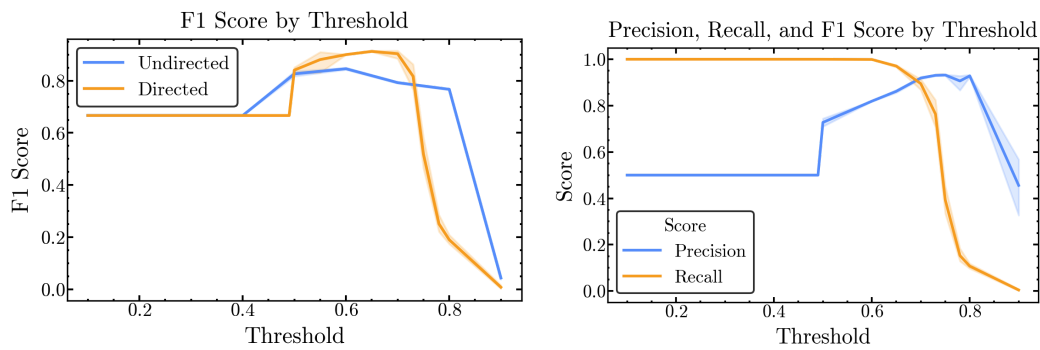


Figure 4.7. Analysis of F1 Score by Classification Threshold for Directed vs. Undirected Models, including the Trade-Off Between Precision and Recall for a directed graph

The precision-recall trends explain this difference. Recall is perfect (1.0) at very low thresholds because the model labels nearly all potential links as positive. However, recall drops sharply after 0.6 and approaches zero beyond 0.8. Precision follows the opposite pattern: it starts low due to many false positives, then steadily increases and exceeds 0.8 around the 0.6–0.7 range. The combination of high precision and still-reasonable recall in this region produces the F1 peak observed for the directed model. Once the threshold climbs past this range, recall collapses rapidly, causing F1 to fall. These dynamics clarify how different thresholds shift model behavior.

- **Around 0.6:** Recall is higher, making this threshold preferable when missing a true link is more costly than predicting a false one.
- **Around 0.7:** Precision and recall are most effectively balanced, yielding the best F1 Score.
- **Around 0.8:** Precision increases further, but recall becomes extremely low, making this setting suitable only when false positives must be minimized.

Interpreting Predictions Relative to the Threshold. If the threshold is set to 0.7 and the model outputs 0.69, the prediction is treated as no link—even though 0.69 exceeds the common default threshold of 0.5. This is because the threshold selected for this specific task defines what counts as a positive prediction. Choosing a stricter threshold (0.7) means the model must produce stronger evidence before classifying a link as present. Thus, a score of 0.69 indicates insufficient confidence under this optimized threshold.

Overall, the analysis shows that a threshold near 0.7 provides the best balance of precision and recall and produces the highest F1 Score, with small adjustments within the 0.6–0.7 region having minimal impact. Performance

deteriorates rapidly outside this range, making it the most reliable operating point for the directed model.

4.2.4 Graph Orientation

Figure 4.8 presents the effect of graph directionality on PR-AUC, precision, and recall across training epochs. The undirected model achieved higher overall PR-AUC, reaching about 0.90 compared to 0.85 for the directed model, which indicates stronger discriminative performance. Precision patterns differed between the two. The undirected model peaked near 0.90 but declined to around 0.77, while the directed model remained steadier at about 0.80. Recall showed the reverse trend. The directed model quickly reached perfect recall and maintained it, whereas the undirected model dipped to around 0.85 before steadily improving to nearly 0.98.

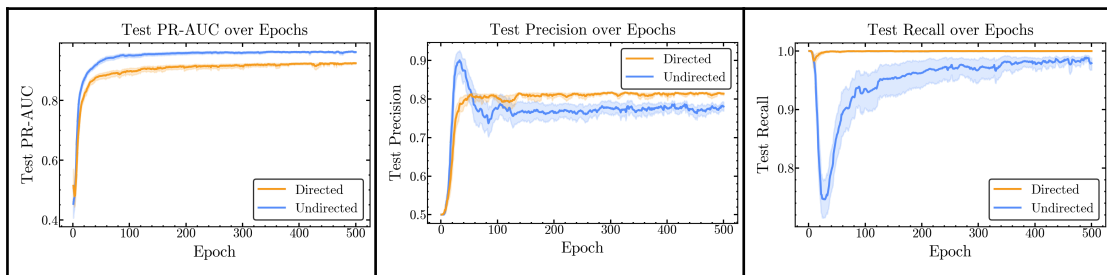


Figure 4.8. Impact of Directional Information (Directed: True vs. False) on Test Performance Metrics (PR-AUC, Precision, and Recall) Across Training Epochs

These findings show a clear trade-off. The undirected model provides higher PR-AUC and a more balanced outcome, but with less stable precision. The directed model ensures perfect recall and slightly stronger precision at convergence, though at the cost of overall balance. When recall is critical, the directed configuration is more suitable, while the undirected configuration is preferable when optimizing the trade-off between precision and recall.

4.2.5 Aggregation Method

Figure 4.9 evaluates different GraphSAGE aggregation methods—median, max, mean, and sum—and shows minimal variation in long-term predictive performance. All four methods rapidly converged to a high and stable Test PR-AUC of approximately 0.90 within the first 100 epochs, with initial variance diminishing quickly. Test F1 Score results were

similarly close across methods. The max, mean, and sum aggregators exhibited a slight but consistent advantage, converging near 0.85 after 400 epochs. The median aggregator, however, displayed noticeable fluctuations throughout training, indicating less stability compared to the other approaches. Overall, these findings suggest that for this dataset and task, the choice of aggregation function has limited impact on final performance, although max, mean, and sum offer marginal improvements in stability and predictive accuracy.

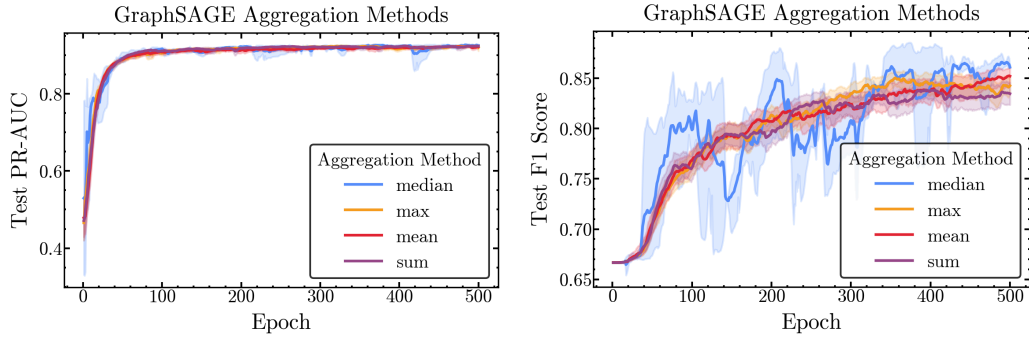


Figure 4.9. Comparison of GraphSAGE Aggregation Methods (Median, Max, Mean, Sum) on Test PR-AUC and F1 Score

4.2.6 Final Best Model Configuration

Table 4.2 summarizes the configuration of the best-performing model based on the results of the ablation study and performance evaluations.

Table 4.2. Best Model Configuration

ASPECT	CONFIGURATION
GNN Architecture	GraphSAGE
Hidden Dimension Size	8
Hidden Layers	2
Classification Threshold	0.7
Graph Orientation	Directed

Aggregation Method	Mean
--------------------	------

The GraphSAGE architecture with mean aggregation achieved the most consistent and reliable performance across metrics. A hidden dimension size of 8 and two hidden layers provided an effective balance between model complexity and generalization. The directed graph orientation captured directional relationships between users and tweets, while a classification threshold of 0.7 yielded the best trade-off between precision and recall. Overall, this configuration demonstrated optimal performance.

4.3 Link Prediction Performance

The link prediction performance of the best model, evaluated using the fine-tuned dataset, is shown in Table 4.3. The model achieved high predictive reliability, with a PR-AUC of 0.905 ± 0.004 , indicating strong capability in distinguishing between positive and negative links even under class imbalance. The accuracy of 0.917 ± 0.003 and the F1 score of 0.923 ± 0.002 further confirm the model's robustness, demonstrating that it effectively captures both structural and textual cues when inferring missing or latent relationships in the network.

Table 4.3. Link Prediction Performance of the Best Configuration using Fine-Tuned Dataset

ACCURACY	PRECISION	RECALL	F1-SCORE	PR-AUC
0.917 ± 0.003	0.862 ± 0.004	0.994 ± 0.002	0.923 ± 0.002	0.905 ± 0.004

Precision (0.862 ± 0.004) reflects the model's ability to generate high-quality positive predictions, while the exceptionally high recall of 0.994 ± 0.002 shows that the model successfully retrieves nearly all true positive links. This is particularly valuable in extremism analysis, where missing a true connection may obscure critical patterns of radicalization or coordinated activity.

Together, these results highlight the effectiveness of integrating BERT HateXplain embeddings with the GraphSAGE-based link prediction framework, showcasing strong generalization and dependable predictive behavior across all metrics.

4.4 Interpretation of Dot-Product Link Scores

The dot product of two vectors provides a measure of directional similarity while incorporating the magnitude of both vectors. Within the context of representation learning, this operation reflects how closely two node embeddings align in both direction and scale. A higher dot-product value indicates stronger similarity between nodes, which in turn suggests a higher likelihood of a link between them. This interpretation aligns with common practices in embedding-based factorization methods used in recommendation systems (Malitesta et al., 2024).

This behavior is illustrated through a simple case study involving two synthetic user profiles presented in Figure 4.10: (1) an unpopular user with zero favourites, followers, and following, and (2) a highly popular user with 100,000 followers and 10,000 favourites and following. The resulting embeddings indicate that the unpopular user receives a sparser representation, whereas the popular user produces a denser embedding.

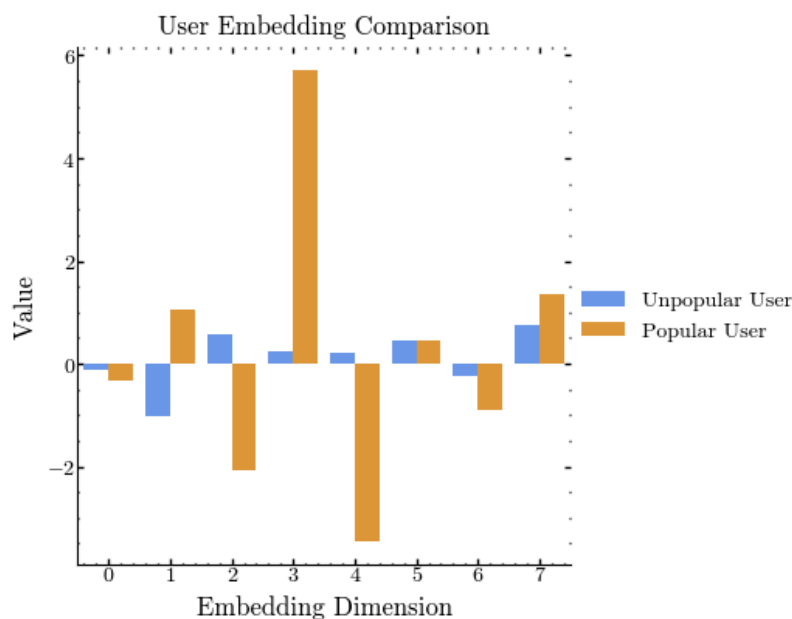


Figure 4.10. Persona Embedding Comparison

Figure 4.11 displays the extremist tweet embeddings associated with the most and least likely user-tweet pairs for both personas. A pair denotes a potential link between a user and a tweet, with its probability derived by passing the dot product of their embeddings through a sigmoid function.

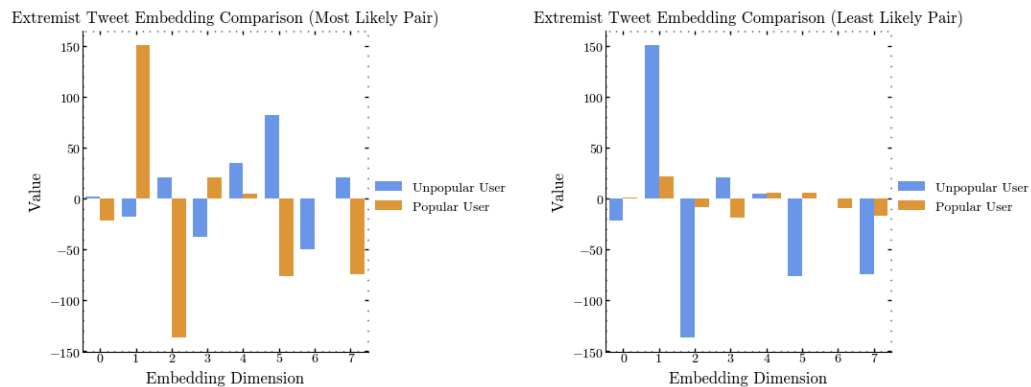


Figure 4.11. Comparison between unpopular and popular users' extreme pairs

The results demonstrate an inverse relationship between user popularity and embedding sparsity: popular users tend to pair with tweets represented by denser embeddings, while unpopular users align more frequently with sparser tweet embeddings. Notably, the tweet associated with the least likely pair for the unpopular user is the same tweet associated with the most likely pair for the popular user. Additional details regarding these pairings are provided in Appendix C.

The sigmoid function outputs normalized values in the range of 0 to 1. Inputs approaching the positive bound yield probabilities near 1, inputs near the negative bound produce probabilities close to 0, and values near 0 correspond to outputs around 0.5, indicating model uncertainty.

Overall, the model shows a consistent pattern: dense user embeddings, characteristic of popular users, tend to produce high link probabilities when paired with dense tweet embeddings. Conversely, sparse user embeddings are more likely to align with sparse tweet representations. This observation reinforces earlier findings that extremist content can attract interaction patterns that do not strictly correlate with tweet-level popularity metrics (see Figure 3.8).

4.5 Application Demo

To demonstrate the practical utility of the proposed framework, a web-based application was developed to allow users to interactively explore the model's predictions. The application provides an accessible interface for analyzing potential extremist content on X/Twitter, bridging the gap between the computational model and end-users such as researchers, analysts, or moderators. This demo serves both as a validation of the model's performance and as a tool to illustrate its interpretability in a real-world setting.

Upon entering the website, users are greeted with the landing page, shown in Figure 4.12, which prominently displays the project title and purpose. Clicking the “Try the Model” button directs users to the inference page, where the core functionality of the model becomes accessible, and users can interactively explore its predictions.

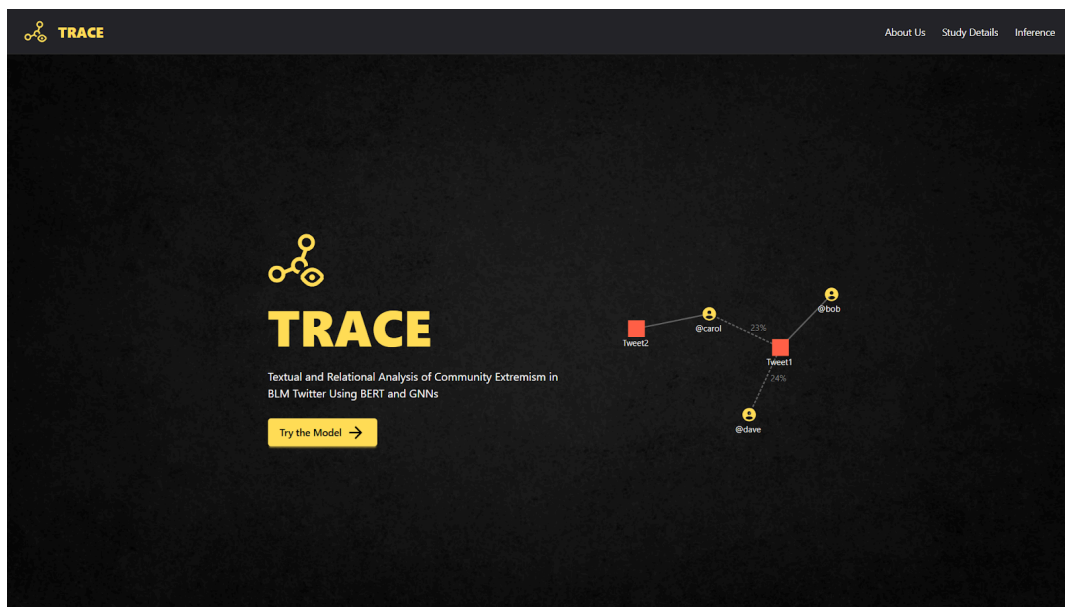
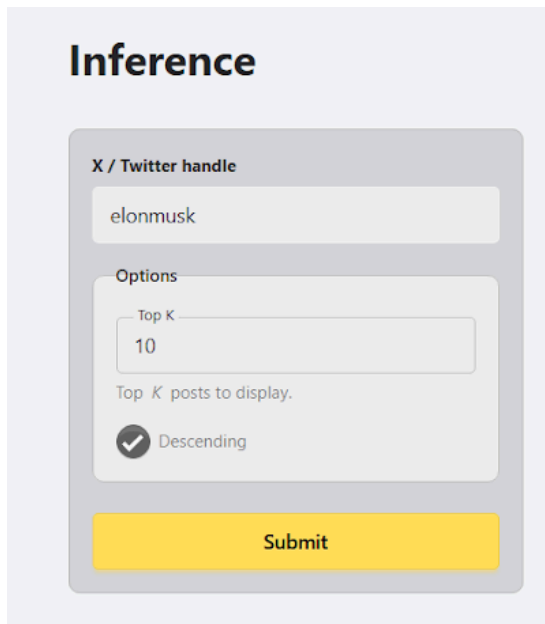


Figure 4.12. Landing Page

On the inference page (Figure 4.13), users can input an X/Twitter handle into the designated field. They can also specify the number of top-k posts they wish to display, allowing for flexibility in the scope of analysis. Additionally, a toggle for ascending or descending order provides control over how posts are ranked, either by probability score or chronological order, enhancing the exploratory capabilities of the interface.



Inference

X / Twitter handle

elonmusk

Options

Top K

10

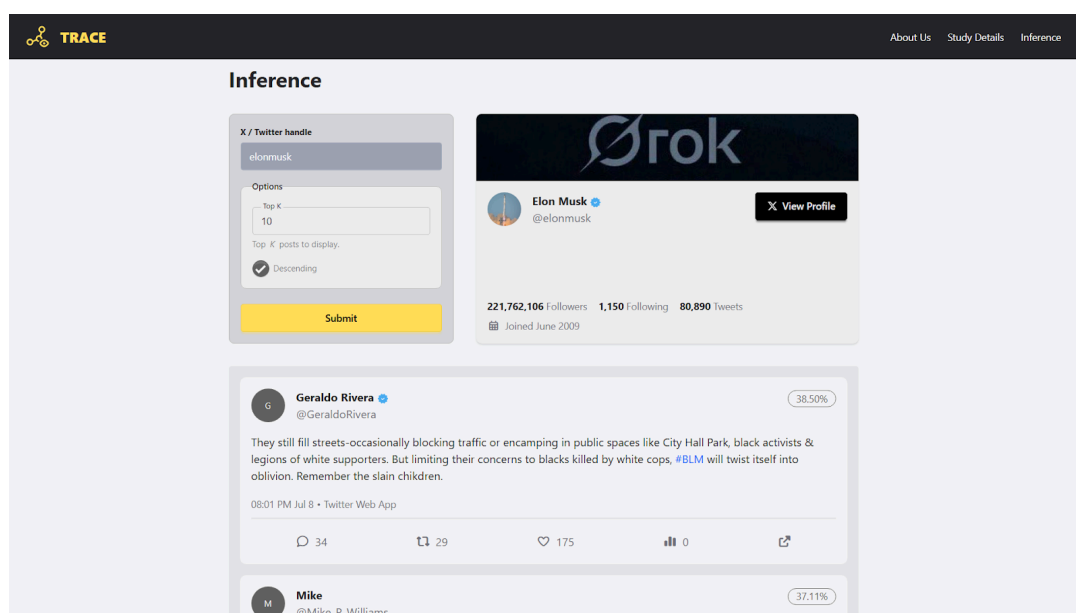
Top K posts to display.

☒ Descending

Submit

Figure 4.13. Inference Panel

After clicking Submit, the selected user's X/Twitter feed is previewed on the right-hand side of the page (Figure 4.14). Below the preview, a list of the top-k posts is displayed according to the chosen ordering. Each post is accompanied by a probability score in the upper-right corner, visually indicating the model's assessment. Posts labeled as extremist by the model are highlighted in red, non-extremist posts in grey, and posts with a 0% probability are highlighted in green for clarity.



TRACE [About Us](#) [Study Details](#) [Inference](#)

Inference

X / Twitter handle

elonmusk

Options

Top K

10

Top K posts to display.

☒ Descending

Submit

Elon Musk [View Profile](#)

@elonmusk

221,762,106 Followers 1,150 Following 80,890 Tweets

Joined June 2009

Geraldo Rivera [View Profile](#)

@GeraldoRivera

38.50%

They still fill streets-occasionally blocking traffic or encamping in public spaces like City Hall Park, black activists & legions of white supporters. But limiting their concerns to blacks killed by white cops, #BLM will twist itself into oblivion. Remember the slain children.

08:01 PM Jul 8 • Twitter Web App

34 29 175 0

Mike [View Profile](#)

@Mike_P_Williams

37.11%

Figure 4.14. Inference Result

This interactive interface provides several practical benefits. By visually encoding the probability of extremism, the demo allows users to quickly identify high-risk content while maintaining transparency in the model's predictions. The inclusion of sorting and filtering options enhances usability, enabling users to focus on specific aspects of the data, such as the most likely extremist posts or chronological patterns of activity. Moreover, the preview of the user's feed alongside the probability scores provides contextual information, helping analysts interpret the predictions in relation to the actual content.

Overall, the application demo not only validates the technical performance of the proposed framework but also emphasizes its interpretability, flexibility, and potential real-world applicability. It demonstrates how computational models can be effectively translated into practical tools for monitoring, research, and educational purposes in the study of online extremism.

Chapter 5

SUMMARY AND CONCLUSION

This chapter presents a comprehensive summary of the study, synthesizing the main findings from the experiments and analyses conducted in previous chapters. It highlights the performance and interpretability of the proposed link prediction framework, which integrates BERT HateXplain embeddings with Graph Neural Networks (GNNs), and identifies the optimal model configuration for detecting relationships in extremist social media networks. The chapter also discusses the implications of the results, interprets key patterns revealed by the model, and outlines the contributions and limitations of the research, providing a cohesive conclusion to the study.

The study began by evaluating the effect of different embedding strategies on model performance. Using the pre-trained BERT HateXplain embeddings as a baseline showed strong performance for the majority classes, with F1 scores of 0.998 for Normal content and 0.922 for Offensive content. However, the minority extremist class achieved only a moderate F1 score of 0.686, likely due to class imbalance. Fine-tuning the embeddings on the domain-specific dataset substantially improved the detection of extremist content, raising the F1 score for this class to 0.787. This result demonstrates that domain adaptation enhances the model's ability to capture nuanced characteristics of extremist discourse, although some improvements may reflect minor overfitting due to oversampling.

An ablation study was conducted to assess the contributions of individual architectural components. Comparisons between GraphSAGE and GCN architectures revealed a clear advantage for GraphSAGE, which consistently outperformed GCN across all evaluation metrics. GraphSAGE achieved stable test accuracy and F1 scores of approximately 0.88 and 0.90, respectively, indicating its effectiveness in aggregating neighbor features and capturing relational and contextual information in noisy or imbalanced graph structures. In contrast, GCN exhibited degraded performance due to oversmoothing.

Further analysis of hyperparameters informed the optimal model configuration. Smaller hidden dimensions (8-16) provided the best balance between precision and recall, while shallow architectures with two hidden layers avoided the oversmoothing issues observed in deeper networks. Directed graph orientation yielded higher recall and slightly stronger precision, and a classification threshold of 0.7 optimized the F1 score. Among aggregation methods, mean aggregation offered the most stable and reliable

results. The final best model combined GraphSAGE architecture, a hidden dimension size of 8, two hidden layers, a classification threshold of 0.7, directed graph orientation, and mean aggregation.

The link prediction evaluation using the fine-tuned dataset confirmed the robustness of the proposed model. It achieved PR-AUC of 0.905 ± 0.004 , accuracy of 0.917 ± 0.003 , F1 score of 0.923 ± 0.002 , precision of 0.862 ± 0.004 , and recall of 0.994 ± 0.002 . The exceptionally high recall is particularly valuable in extremism analysis, where missing connections could obscure critical patterns of coordination or radicalization.

However, it is important to interpret these link prediction results with caution. The fine-tuned BERT embeddings achieved only a moderate F1 score of 0.787 for the extremist class, indicating that the model's representation of extremist content is imperfect. Consequently, high-probability links predicted by the model should not be taken as definitive evidence of relationships between users and tweets. Rather, the link prediction scores should be viewed as indicative of potential connections, capturing patterns in the data that warrant further qualitative or contextual investigation.

Interpretation of the model outputs, based on the dot product used to measure link probability, revealed consistent patterns. High link probabilities occurred when dense user embeddings paired with dense tweet embeddings, while low probabilities occurred when sparse embeddings aligned. This indicates that the model effectively captures subtle semantic and relational cues between users and tweets, extending beyond simple measures of popularity or activity.

In conclusion, the study demonstrates that integrating BERT HateXplain embeddings with a GraphSAGE-based link prediction framework provides a robust and reliable method for inferring latent relationships in extremist social media networks. The final model achieved high predictive reliability, generalization, and interpretability, but caution must be exercised when interpreting the results, particularly for links involving extremist content. Overall, this research contributes a scalable and effective framework for analyzing complex social networks and extracting actionable insights from text and relational data, while also highlighting the need for careful contextual validation of predicted connections.

Chapter 6 RECOMMENDATIONS

6.1 Recommendations

Based on the findings and insights from this study, several recommendations are proposed to guide future research and practical applications. These recommendations aim to enhance the detection of extremist content, improve the robustness and interpretability of link prediction models, and ensure ethical and responsible deployment in real-world social media environments. They address both methodological improvements and practical considerations, offering a roadmap for advancing computational approaches to extremism analysis.

Extended Fine-tuning and Larger Annotated Datasets. Future studies should explore more extensive fine-tuning of BERT HateXplain, such as unfreezing additional transformer layers, adding auxiliary classification heads, or employing advanced regularization techniques. Expanding the size and diversity of annotated datasets would improve the semantic precision of embeddings, particularly for minority classes such as extremist content, which remain challenging to detect accurately.

Exploration of Advanced and Dynamic GNN Architectures. While GraphSAGE provided strong performance in this study, future research could investigate more expressive architectures, including Graph Attention Networks (GAT), Heterogeneous Graph Neural Networks (HetGNN), or temporal/dynamic graph models. These approaches could better capture evolving social interactions and the spread of extremist narratives over time, providing a more detailed and temporally aware understanding of network dynamics.

Incorporation of Psychological and Behavioral Factors. Integrating psychological or behavioral theories—such as social identity theory, balance theory, or emotional contagion—into graph-based learning could enrich model interpretability. Combining structural graph learning with behavioral insights may help explain the underlying motivations behind extremist or radical interactions, bridging computational models with social science perspectives.

Caution in Interpreting Link Prediction Results. Given that the fine-tuned BERT embeddings achieved a moderate F1 score of 0.787 for the extremist class, predictions from the link prediction framework should not be taken literally as definitive relationships. Future applications should consider link probabilities as indicative of potential connections and validate high-probability links using qualitative, contextual, or expert-driven analyses. Incorporating uncertainty estimation or probabilistic thresholds may improve the practical reliability of predictions.

Bias, Fairness, and Ethical Considerations. Even with interpretable embeddings like HateXplain, careful attention must be paid to algorithmic fairness. Future work should implement bias detection and mitigation strategies to avoid misclassifying marginalized groups as extremists. Ethical oversight and compliance with data privacy standards are essential for responsible deployment.

Real-world Deployment and Multimodal Integration. The pipeline developed in this study can be adapted for social media monitoring or moderation systems. Real-world implementation requires considerations for computational efficiency, scalability, and collaboration with social scientists, policy experts, and ethicists. Extending the framework to incorporate multimodal data—such as images, videos, or memes—and cross-platform interactions would enhance detection of coordinated extremist activity and improve overall robustness.

6.2 Future Work

Future research should focus on improving the accuracy and reliability of extremist content detection, particularly for minority classes. Approaches could include semi-supervised or self-supervised learning to leverage large amounts of unlabeled social media data, active learning to prioritize annotation of ambiguous or high-risk content, and ensemble methods that combine multiple embeddings or GNN architectures.

In addition, developing uncertainty-aware link prediction frameworks would allow the model to quantify confidence in predicted links, helping practitioners distinguish between highly probable connections and those requiring further validation. Integrating temporal modeling and cross-platform data analysis could capture the dynamic nature of online extremist networks, while multimodal fusion (text, images, videos, and memes) would enhance the system’s ability to detect coordinated or multi-faceted radicalization strategies.

Finally, incorporating ongoing ethical evaluation and human-in-the-loop verification will be crucial for deploying these models responsibly, ensuring that predictions support informed decision-making without unintended harms or biases.

6.3 Closing Statement

This research demonstrates the potential of integrating contextualized language embeddings with graph neural networks to capture relational and semantic cues within extremist social media networks. While fine-tuning BERT provided moderate gains, the combination of pre-trained HateXplain embeddings with GraphSAGE proved to be both powerful and interpretable. These findings provide a foundation for future advancements in computational extremism analysis, offering tools for detecting, understanding, and mitigating the spread of extremist content online – while emphasizing the importance of careful interpretation, ethical deployment, and continued methodological refinement.

REFERENCES

- Alatawi, H. S., Alhothali, A. M., & Moria, K. M. (2021). Detecting White Supremacist Hate Speech using Domain Specific Word Embedding with Deep Learning and BERT. arXiv.org. <https://arxiv.org/abs/2010.00357>
- Alhosseini, S., & Meinel, C. (2022). Predicting User Engagements using Graph Neural Networks on Online Social Networks. <https://doi.org/10.1109/HPCC-DSS-SmartCity-DependSys57074.2022.00339>
- Alizadeh, M., Shapiro, J. N., Buntain, C., & Tucker, J. A. (2020). Content-based features predict social media influence operations. *Science Advances*, 6(30). <https://doi.org/10.1126/sciadv.abb5824>
- Apollonio, N., Franciosa, P. G., & Santoni, D. (2022). A novel method for assessing and measuring homophily in networks through second-order statistics. *Scientific Reports*, 12(1). <https://doi.org/10.1038/s41598-022-12710-7>
- Arazzi, M., Cotogni, M., Nocera, A., & Virgili, L. (2023). Predicting Tweet Engagement with Graph Neural Networks. In *Proceedings of the 2023 ACM International Conference on Multimedia Retrieval (ICMR '23)* (pp. 172–180). Association for Computing Machinery. <https://doi.org/10.1145/3591106.3592294>
- Asikainen, A., Iñiguez, G., Ureña-Carrión, J., Kaski, K., & Kivelä, M. (2020). Cumulative effects of triadic closure and homophily in social networks. *Science Advances*, 6(19). <https://doi.org/10.1126/sciadv.aax7310>
- Atske, S., & Atske, S. (2023). #BlackLivesMatter turns 10. Pew Research Center. <https://www.pewresearch.org/internet/2023/06/29/blacklivesmatter-turns-10>
- Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. B. F., Lee, J., Mann, M., Merhout, F., & Volfovsky, A. (2018). Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37), 9216–9221. <https://doi.org/10.1073/pnas.1804840115>
- Benesch, S., Buerger, C., Glavinic, T., & Manion, S. (2018). Dangerous speech: A practical guide. Dangerous Speech Project. <https://dangerousspeech.org/guide/>

- Bermingham, A., Conway, M., McInerney, L., O'Hare, N., & Smeaton, A. F. (2009). Combining social network analysis and sentiment analysis to explore the potential for online radicalisation. *Proceedings of the 2009 International Conference on Advances in Social Network Analysis and Mining*, 231-236. <https://doi.org/10.1109/ASONAM.2009.31>
- Bonilla, Y., & Rosa, J. (2015). #Ferguson: Digital protest, hashtag ethnography, and the racial politics of social media in the United States. *American Ethnologist*, 42(1), 4-17. <https://doi.org/10.1111/amet.12112>
- Borum, R. (2011). Radicalization into violent extremism I: A review of social science theories. *Journal of Strategic Security*, 4(4), 7-36. <https://doi.org/10.5038/1944-0472.4.4.1>
- Boutyline, A., & Willer, R. (2016). The social structure of political echo chambers: Variation in ideological homophily in online networks. *Political Psychology*, 38(3), 551-569. <https://doi.org/10.1111/pops.12337>
- Brady, W. J., Crockett, M. J., & Van Bavel, J. J. (2020). The MAD Model of Moral Contagion: The role of motivation, attention, and design in the spread of moralized content online. *Perspectives on Psychological Science*, 15(4), 978-1010. <https://doi.org/10.1177/1745691620917336>
- Burnap, P., & Williams, M. L. (2016). Us and them: identifying cyber hate on Twitter across multiple protected characteristics. *EPJ Data Science*, 5(1). <https://doi.org/10.1140/epjds/s13688-016-0072-6>
- Châtel, B. D. L., Quax, R., Peeters, G., Corten, R., Rikkert, M. G. M. O., & Vasconcelos, V. V. (2025). Homophily and social influence as mechanisms of loneliness clustering in social networks. *Scientific Reports*, 15(1). <https://doi.org/10.1038/s41598-025-99057-x>
- Chen, H. (2018). Exploring extremist cognition: Linguistic patterns in online extremist communities. *Terrorism and Political Violence*, 30(4), 670-692. <https://doi.org/10.1080/09546553.2018.1442329>
- Chen, T., Zhou, K., Duan, K., Zheng, W., Wang, P., Hu, X., & Wang, Z. (2024). Bag of tricks for training deeper graph neural networks: A comprehensive benchmark study. *arXiv*. <https://arxiv.org/abs/2108.10521>
- Cohen, K., Johansson, F., Kaati, L., & Mork, J. C. (2014). Detecting linguistic markers for radical violence in social media. *Terrorism and Political Violence*, 26(1), 246-256. <https://doi.org/10.1080/09546553.2014.849948>
- Denemark, J. (2023). Strengthening the european union by regulating the digital single market. *Auc Iuridica*, 69(2), 107-123. <https://doi.org/10.14712/23366478.2023.18>

- Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv.org. <https://arxiv.org/abs/1810.04805v2>
- Devyatkin, D., Smirnov, I., & Sochenkov, I. (2017). Exploring linguistic features for extremist texts detection (on the material of Russian-speaking illegal texts). 2017 IEEE International Conference on Intelligence and Security Informatics (ISI), 188–190. <https://doi.org/10.1109/ISI.2017.8004907>
- Dreißigacker, A., Müller, P., Isenhardt, A. et al. (2024). Online hate speech victimization: consequences for victims' feelings of insecurity. *Crime Science*, 13, 4. <https://doi.org/10.1186/s40163-024-00204-y>
- Evtushenko, A., & Kleinberg, J. (2021). The paradox of second-order homophily in networks. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-92719-6>
- Ferrara, E., Wang, W., Varol, O., Flammini, A., & Galstyan, A. (2016). Predicting online extremism, content adopters, and interaction reciprocity. In *Lecture notes in computer science* (pp. 22–39). https://doi.org/10.1007/978-3-319-47874-6_3
- Freelon, D., McIlwain, C. D., & Clark, M. D. (2016). Beyond the hashtags: #Ferguson, #Blacklivesmatter, and the online struggle for offline justice. Center for Media & Social Impact, American University. <https://doi.org/10.2139/ssrn.2747066>
- Gallagher, R. J., Frank, M. R., Mitchell, L., Schwartz, A. J., Reagan, A. J., Danforth, C. M., & Dodds, P. S. (2021). Generalized word shift graphs: A method for visualizing and explaining pairwise comparisons between texts. *EPJ Data Science*, 10(1), 1-41. <https://doi.org/10.1140/epjds/s13688-021-00260-3>
- González-Bailón, S., & Wang, N. (2015). Networked discontent: The anatomy of protest campaigns in social media. *Social Networks*, 44, 95–104. <https://doi.org/10.1016/j.socnet.2015.07.003>
- Hamilton, W. L., Ying, R., & Leskovec, J. (2018). Inductive representation learning on large graphs. arXiv.org. <https://arxiv.org/abs/1706.02216v4>
- Hawkins, D. S. (2023). “When you Search a #Hashtag, it Feels Like You’re Searching for Death:” Black Twitter and Communication About Police Brutality Within the Black Community. *Social Media + Society*, 9(2). <https://doi.org/10.1177/20563051231179705>
- Holbrook, D. (2014). The Al-Qaeda doctrine: Training texts and language. *Terrorism and Political Violence*, 26(5), 823-844. <https://doi.org/10.1080/09546553.2013.828604>

- Jamil, M. L., Pais, S., Cordeiro, J., & Dias, G. (2022). Detection of extreme sentiments on social networks with BERT. *Social Network Analysis and Mining*, 12(1). <https://doi.org/10.1007/s13278-022-00882-z>
- Khokhlovskaya, O. G., & Chernyavskaya, K. E. (2024). Linguistic markers of extremism in Islamist texts of the religious genre "sermon": Methodological aspect. *Bulletin of Chelyabinsk State University*. <https://doi.org/10.47475/1994-2796-2024-487-5-135-142>
- Kipf, T. N., & Welling, M. (2017). Semi-Supervised Classification with Graph Convolutional Networks. *arXiv.org*. <https://arxiv.org/abs/1609.02907v4>
- Kumar, S., Beniwal, R., Singh, S., & Gupta, V. (2019). Predicting Link Sign in Online Social Networks based on Social Psychology Theory and Machine Learning Techniques. <https://doi.org/10.1109/ICCICC46617.2019.9146087>
- Li, Z., Xin, S., Jiao, Y., Pan, X., Zou, P., Meng, X., Yao, C., & Bu, J. (2020). Hierarchical bipartite Graph neural Networks: towards Large-Scale E-commerce Applications. <https://doi.org/10.1109/ICDE48307.2020.00149>
- Malitesta, D., Mancino, A. C. M., Minervini, P., & Tommaso, D. N. (2024). Dot Product is All You Need: Bridging the Gap Between Item Recommendation and Link Prediction. *arXiv.org*. <https://arxiv.org/abs/2409.07433>
- Mathew, B., Saha, P., Yimam, S. M., Biemann, C., Goyal, P., & Mukherjee, A. (2021). HateXplain: A benchmark dataset for explainable hate speech detection. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(17), 14867-14875. <https://doi.org/10.1609/aaai.v35i17.17745>
- Müller, K., & Schwarz, C. (2021). Fanning the flames of hate: Social media and hate crime. *Journal of the European Economic Association*, 19(4), 2131-2167. <https://doi.org/10.1093/jeea/jvaa045>
- Oniani, D., Chandrasekar, P., Sivarajkumar, S., & Wang, Y. (2023). Few-shot learning for clinical natural language processing using Siamese neural networks: Algorithm development and validation study. *JMIR AI*, 2, e44293. <https://doi.org/10.2196/44293>
- Pariser, E. (2011). *The filter bubble: What the Internet is hiding from you*. Penguin UK.
- Phadke, S., Samory, M., & Mitra, T. (2021). What makes people join conspiracy communities? Role of social factors in conspiracy engagement. *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW3), 1-30. <https://doi.org/10.1145/3432922>

- Phadke, S., Seiler, J., Mitra, T., Garimella, K., Costello, M., & E.Hawdon, J. (2021). Addressing challenges and opportunities in online extremism research: an interdisciplinary perspective., 356-359. <https://doi.org/10.1145/3462204.3481722>
- Praznik, L., Qudar, M. M. A., Mendhe, C., Srivastava, G., & Mago, V. K. (2021). Analysis of link prediction algorithms in hashtag graphs. https://doi.org/10.1007/978-3-030-67044-3_11
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. <https://doi.org/10.18653/v1/d19-1410>
- Ribeiro, M. H., Ottoni, R., West, R., Almeida, V. A., & Meira Jr, W. (2020). Auditing radicalization pathways on YouTube. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, 131-141. <https://doi.org/10.1145/3351095.3372879>
- Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. (2020). Temporal Graph networks for deep learning on dynamic graphs. arXiv.org. <https://arxiv.org/abs/2006.10637>
- Sadeque, F., & Bethard, S. (2019). Predicting engagement in online social networks: Challenges and opportunities. arXiv.org. <https://arxiv.org/abs/1907.05442>
- Schmid, A. P. (2013). Radicalisation, de-radicalisation, counter-radicalisation: A conceptual discussion and literature review. The International Centre for Counter-Terrorism, 4(2), 1-91. <https://doi.org/10.19165/2013.1.02>
- Shahin, A. F. (2024). Linguistic Stylistics of Extremist Discourse in the Digital Realm: Analyzing ISIS's rhetoric. Ansaq Journal, 8(1). <https://doi.org/10.29117/ansaq.2024.0198>
- Soler-Company, J., & Wanner, L. (2019). Automatic classification and linguistic analysis of extremist online material. In Lecture notes in computer science (pp. 577-582). https://doi.org/10.1007/978-3-030-05716-9_49
- Subramaniam, A., Mehra, A., & Kundu, S. (2022). Exploring Hate Speech Detection with HateXplain and BERT. arXiv.org. <https://arxiv.org/abs/2208.04489>
- Torregrosa, J., Bello-Orgaz, G., Martínez-Cámara, E., Del Ser, J., & Camacho, D. (2022). A survey on extremism analysis using natural language processing: definitions, literature review, trends and challenges. Journal of Ambient Intelligence and Humanized Computing, 14(8), 9869-9905. <https://doi.org/10.1007/s12652-021-03658-z>

- Torregrosa, J., Panizo-Lledot, Á., Bello-Orgaz, G., & Camacho, D. (2020). Analyzing the relationship between relevance and extremist discourse in an alt-right network on Twitter. *Social Network Analysis and Mining*, 10(1). <https://doi.org/10.1007/s13278-020-00676-1>
- Tufekci, Z. (2017). *Twitter and tear gas: The power and fragility of networked protest*. Yale University Press. <https://doi.org/10.25969/mediarep/14848>
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. arXiv.org. <https://arxiv.org/abs/1710.10903v3>
- Vergani, M., & Bliuc, A. M. (2015). The evolution of the ISIS' language: A quantitative analysis of the language of the first year of Dabiq magazine. *Sicurezza, Terrorismo e Società*, 2(2), 7-20. <https://doi.org/10.13137/2421-4604/11596>
- Wang, Y. (2022). Friend Recommendation using GraphSAGE - Stanford CS224W: Machine Learning with Graphs - Medium. Medium. <https://medium.com/stanford-cs224w/friend-recommendation-using-graphsage-ffcda2aaf8d6>
- Wolfowicz, M., Weisburd, D., & Hasisi, B. (2021). Examining the interactive effects of the filter bubble and the echo chamber on radicalization. *Journal of Experimental Criminology*, 19(1), 119-141. <https://doi.org/10.1007/s11292-021-09471-0>
- Wu, L., Cui, P., Pei, J., Zhao, L., & Song, L. (2024). A review of graph neural networks: Concepts, architectures, techniques, challenges, datasets, applications, and future directions. *Journal of Big Data*, 11(1), Article 16. <https://doi.org/10.1186/s40537-023-00876-4>
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Yu, P. S. (2020). A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(1), 4-24. <https://doi.org/10.1109/tnnls.2020.2978386>
- Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., & Leskovec, J. (2018). Graph Convolutional Neural Networks for Web-Scale Recommender Systems. arXiv.org. <https://arxiv.org/abs/1806.01973v1>
- Zeng, H., Zhou, H., Srivastava, A., Kannan, R., & Prasanna, V. K. (2020). GraphSAINT: Graph Sampling Based Inductive Learning Method. ICLR 2020. <https://arxiv.org/abs/1907.04931v4>
- Zhang, C., Zhou, Z., Peng, X., & Xu, K. (2023). DoubleH: Twitter user stance detection via bipartite graph neural networks. arXiv.org. <https://arxiv.org/abs/2301.08774>

APPENDIX A

Average Interaction Metrics by Tweet Class

This appendix presents the average engagement metrics across tweet classes derived from the labeled dataset. The results in Table A.1 show notable variation in user interactions such as favorites, replies, retweets, quotes, and bookmarks among the three classes.

Table A.1. Average Interaction Counts of Tweet Features

CLASS	FAVORITE	REPLY	RETWEET	QUOTE	BOOKMARK
0	75.04	9.72	19.31	4.09	1.06
1	198.98	8.70	74.78	6.66	2.55
2	88.311	5.60	22.92	2.83	1.03

Tweets under Class 1 (Normal) received the highest average counts in favorites and retweets, suggesting that these posts tend to be more widely liked and shared. Class 0 (Extremist) tweets, while less popular in terms of likes and shares, recorded higher reply interactions, indicating greater conversational engagement. Meanwhile, Class 2 (Offensive) tweets demonstrated comparatively lower interaction levels across all engagement types, suggesting neutral or less engaging content overall. These findings support the trends discussed in Chapter 3, where user behavior and tweet engagement were found to differ substantially by class.

APPENDIX B

Deployed System Access

To demonstrate the practical implementation and usability of the proposed models, a production-ready prototype was deployed as part of MVP 3.0. The system architecture integrates a FastAPI-based backend inference engine with a Svelte frontend, deployed through Cloudflare Workers to ensure scalability, low latency, and high availability. The deployed application is accessible at the following link: <https://capstone.ustp.party>

The online platform allows users to input Twitter handles for automated analysis of user activity. The system retrieves the corresponding tweets and computes the link probability between users and posts based on the trained model's predictions. Users may also adjust the Top-K parameter, which determines the number of posts displayed for each account, enabling customized exploration of user behavior. The interface also presents model performance metrics, classification summaries, and detailed information about the study and its authors, promoting transparency and interpretability. An image of the web interface is shown in Figure B.1.

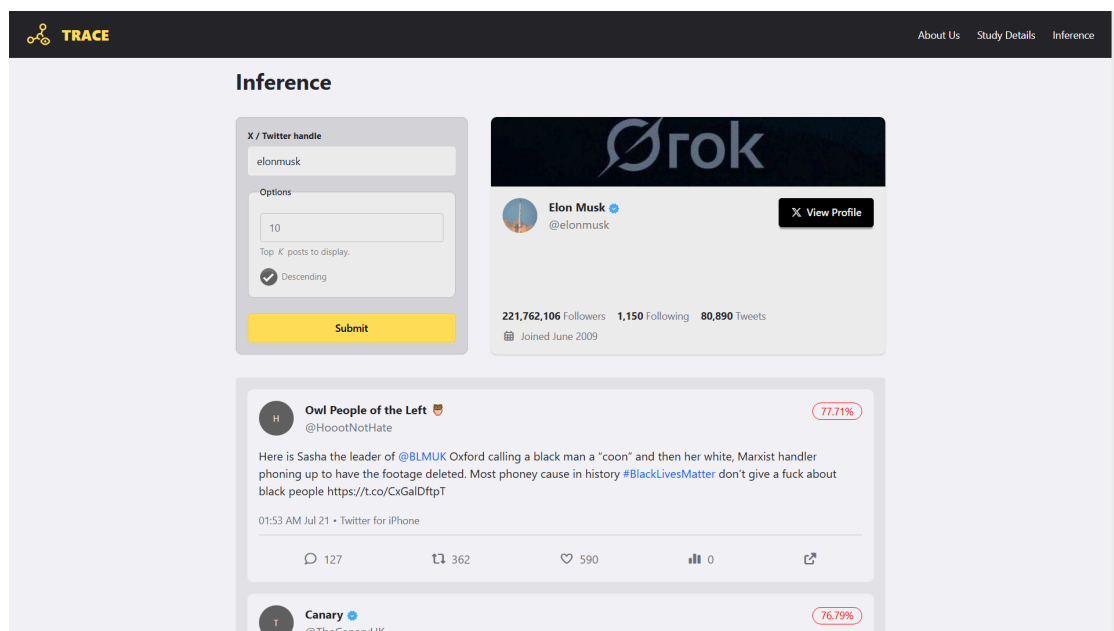


Figure B.1. Screen capture of the deployed web interface showing user handle input, Top-K configuration, and interactive social graph visualization.

This web-based deployment ensures accessibility for both technical and

non-technical stakeholders, allowing them to explore and evaluate the system's analytical capabilities without requiring local installation or configuration.

The project's source code and supplementary documentation are publicly available at the following repositories:

- GitHub Repository:
<https://github.com/iragca/capstone-project-2>
- Project Documentation:
<https://iragca.github.io/capstone-project-2/>

APPENDIX C

Persona Experiment Results

The results of the persona experiment reveal an inverse relationship between user popularity and embedding sparsity. Popular users tend to pair with tweets represented by denser embeddings, whereas less popular users more frequently align with sparser tweet embeddings. As shown in Table C.1, the most likely extremist tweet pairings indicate minimal engagement for popular users, while unpopular users engage slightly more, though the overall interaction remains low.

Table C.1. Most likely paired extremist tweets for popular and unpopular users.

USER	FAVORITE COUNT	REPLY COUNT	QUOTE COUNT	SOURCE	IS EXTREMIST
Popular	0	0	0	0	0
Unpopular	1	1	0	0	0

In contrast, Table C.2 presents the least likely extremist tweet pairings, where the tweet associated with the highest engagement for popular users is completely absent in the interactions of unpopular users. This reinforces the observation that user popularity strongly influences the system’s pairing behavior and reflects social engagement patterns captured by the model.

Table C.2. Least likely paired extremist tweets for popular and unpopular users.

USER	FAVORITE COUNT	REPLY COUNT	QUOTE COUNT	SOURCE	IS EXTREMIST
Popular	2473	259	56	0	0
Unpopular	0	0	0	0	0

These findings suggest that the model not only differentiates tweet characteristics but also implicitly captures user engagement dynamics. Popular users tend to be paired with highly interactive tweets, represented by dense embeddings, while unpopular users are paired with sparser tweets. The fact that the same tweet can be the most likely pairing for a popular user and the least likely for an unpopular user highlights how social influence affects the system's predictions. Such insights can inform strategies for monitoring and analyzing extremist content, accounting for both tweet characteristics and user popularity.